Screen or Barrier? Science of Reading Certification Exams and Teacher Supply in Texas. *

Andrew Avitabile
University of Virginia
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Abstract

Twenty states now require prospective teachers to pass a standalone Science of Reading exam. These exams restrict teacher supply as a *barrier*, and whether they are worth that cost depends on whether they also *screen* on teaching quality. I study Texas' exam, required since 2021 for new elementary and middle-grades candidates, using administrative data linking exam attempts, certificates, employment, and student achievement. In a regression discontinuity at the passing threshold, narrowly failing lowers the probability of teaching the following year by 6.5 percentage points, driven by alternatively certified candidates. In a difference-in-differences design comparing affected to unaffected certification fields within the same preparation programs, new certifications in affected fields fell roughly 18 percent and have not recovered five years later. Against these costs, exam scores predict student reading achievement two to four times more strongly than general pedagogy scores, and first-attempt failers who go on to teach have reading value-added 0.016 to 0.021 standard deviations below narrow passers. The exam is thus a rare pre-hire predictor of teacher effectiveness, and lowering the threshold would likely admit less effective teachers. Structured Science of Reading coursework at preparation programs raised first-attempt pass rates about 4 percentage points, suggesting states can lower the barrier without lowering the standard.

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1 Introduction

Teacher certification exams occupy an uneasy position in education policy. They create a *barrier* to teacher supply (M. Kleiner, 2006), but they may also be a *screen* if they filter candidates on knowledge relevant to student learning (Stiglitz, 1975). Whether an exam is worth its cost to teacher supply depends on which dominates, and for two decades the empirical evidence has pointed one way. More stringent requirements reduce teacher supply without improving quality (Angrist & Guryan, 2008; Deneault et al., 2025), passing thresholds do a poor job of separating more from less effective candidates (Goldhaber, 2007; Orellana & Winters, 2025), and exam scores are, at best, weakly predictive of teacher effectiveness (Goldhaber et al., 2017a, 2017b; T. J. Kane et al., 2008; Rockoff et al., 2011). The conclusion drawn from this evidence is that states should lower barriers to entry such as certification exams and instead reward teachers who raise student achievement (Hanushek & Rivkin, 2007).

Following this evidence some states have dropped testing requirements (NJ State Policy Lab, 2025; Swisher, 2022), but two recent developments have pushed others in the opposite direction. First, teachers who entered through relaxed pandemic-era pathways proved somewhat less effective than their traditionally certified peers (Backes et al., 2024; J. J. Kirksey, 2024). Second, roughly 40 percent of fourth graders now score below basic in reading on the National Assessment of Educational Progress (NAEP), the highest share since 2002 (National Center for Education Statistics, 2025), and a large body of research points to a specific treatment. Effective literacy instruction requires explicit, systematic teaching of foundational skills (Castles et al., 2018; Seidenberg, 2017)—called the Science of Reading (SOR)—but educator preparation programs (EPPs) often do not provide prospective teachers with these skills (Ellis et al., 2023). In response, twenty states now require a standalone SOR exam as a condition of certification (National Council on Teacher Quality, 2024).

Despite this trend, states lack evidence on how these exams affect the teacher pipeline. In this paper I ask two questions. Are SOR certification exams merely a barrier to supply, or do they also screen on teacher effectiveness? And what can policymakers do to help candidates meet the standard, short of removing it?

I answer these questions in the context of the Texas Examinations of Educator Standards (TExES) SOR exam, announced in June 2019 and required beginning January 1, 2021 of all new candidates in early childhood through grade 6 and grades 4–8 core subjects and English language arts. The data for this study cover the universe of Texas certification exams, certificates, and public school employment, and are linked to student achievement records. To measure the exam as a barrier, I use a regression discontinuity at the passing threshold that identifies the effect of narrowly failing on certification and employment, and a difference-in-differences design that compares the flow of new certifications in fields that began requiring the exam to fields within the same preparation programs that did not. To measure the exam as a screen, I explore the extent to which SOR scores predict student achievement and whether the candidates the exam excludes are likely to be significantly less effective than the ones it admits. Finally, I ask whether Texas Reading Academies (TRAs), state-developed SOR coursework delivered by EPPs, can raise pass rates on SOR certification exams.

The SOR exam is a significant barrier to teacher supply. Narrowly failing reduces the probability of earning a teaching certificate within six months by 9.2 percentage points and of working as a teacher the following October by 6.5 percentage points. The employment cost depends on where the exam falls in a candidate's route. Traditional candidates take it at the end of their preparation, and for them a narrow failure delays certification but not subsequent employment. Alternatively certified candidates must pass before their program can place them on the paid internship the route requires, so a narrow failure for those on this route lowers the probability of teaching the next year by 12 percentage points. At the market level, new certifications in affected fields spiked ahead of the mandate as candidates rushed to certify under the old rules, then fell and have not recovered five years later. Netting out the pre-announcement differential trend, certifications in affected fields sit roughly 18 percent below comparison fields within the same EPP. I also find no evidence of substitution, as candidates who narrowly fail are no more likely to certify in unaffected fields, and post-mandate cohorts certify more slowly in affected fields without certifying faster elsewhere.

On the screening side, I ask whether the exam provides a pre-hire signal of teaching quality and how the candidates the passing threshold excludes compare to the ones it admits.

In student-level value-added models, students assigned to a teacher who scored one standard deviation higher on the SOR exam score 0.013 standard deviations higher in reading within the same school, under a tenth of a standard deviation in teacher effectiveness in reading. This is two to four times the gain associated with the general pedagogy exam that all candidates take, and it is comparable to the strongest licensure-test associations reported in other states (Clotfelter et al., 2007; Goldhaber, 2007). Entering both scores in the same model, the SOR coefficient remains essentially unchanged while the pedagogy coefficient falls to zero. Given how difficult it has proven to predict teacher effectiveness before hire (Clotfelter et al., 2007; Cowan et al., 2023; T. Kane & Staiger, 2008; Staiger & Rockoff, 2010), this contrast suggests the SOR exam captures something about teaching reading that the general pedagogy exam does not. I also find evidence that the threshold excludes candidates who are less effective than the ones it admits. Comparing candidates who passed by a narrow margin to first-attempt failers who went on to teach, and progressively widening the band of failers included below the cutoff, I find that failers have reading value-added 0.016 to 0.021 standard deviations lower. This gap is likely understated, since first-attempt failers who persist into the classroom have higher SAT scores and family incomes than failers who do not. I interpret this as evidence that lowering the threshold would admit teachers who are, on average, less effective at raising reading achievement than those it currently admits.

Taken together, these results put states in a bind. A state deciding whether to adopt or keep a Science of Reading exam is choosing between pre-hire information about teaching quality and a considerable cost to teacher supply. All at a moment when reading achievement and interest in teaching are at or near their lowest levels in decades (Bartanen & Kwok, 2023; Kraft & Lyon, 2024; National Center for Education Statistics, 2025). This need not be the only choice if states could keep the standard and help candidates meet it, but the licensure literature offers almost no evidence on this option. Here I provide preliminary evidence on it using Texas Reading Academies, structured SOR coursework developed by the state, which ten EPPs began delivering to preservice candidates in 2024–25. First-attempt SOR pass rates at these programs rose about 4 percentage points relative to programs that never adopted. This evidence is early, as the adopting programs are few, pass rates on the pedagogy exam also rose in the adoption year, and a second cohort of adopters fails a pre-trend test. But

the results suggest that the supply cost of an exam is not fixed, and that a state can lower it without lowering the bar. As more states add SOR requirements, how to help candidates pass deserves as much attention as whether to test.

2 Background

2.1 Conceptual Framework: Licensure as Screen and Barrier

This paper draws on three related economic traditions. The economics of occupational licensing (M. Kleiner, 2006; M. M. Kleiner & Krueger, 2013) treats licensure as a labor market institution performing two functions simultaneously. First, a *screening* function that filters candidates on minimum competency and second a *barrier* function that imposes costs (e.g., fees, preparation time, retake delays, and foregone earnings) on prospective entrants. The welfare consequences of a licensure requirement depend on the balance between these two. If an exam screens effectively, its supply costs may be offset by quality gains. If it imposes barriers without meaningful screening, it restricts supply without compensating benefits.

Whether an exam screens effectively depends on what its scores measure. The classical distinction between *screening* (Stiglitz, 1975) and *signaling* (Spence, 1973) is useful here. A screen identifies differences in candidate knowledge relevant to performance, while a pure signal produces a credential whose value derives from the cost of obtaining it rather than what it certifies. Together, these traditions generate three testable predictions that organize the empirical work. First, if the exam functions as a binding barrier, failing should delay certification and reduce subsequent employment, with larger effects for candidates already at the margin of the labor market. Second, if the exam screens rather than signals, scores should predict student reading achievement and stop less effective candidates from entering the profession. Third, if the exam screens and barriers stem from inadequate preparation rather than fixed differences in candidate ability, then structured preparation that builds this knowledge is an investment in human capital (Becker, 1964) and should improve exam performance. If the exam is instead closer to a pure signal, such preparation may help

candidates pass without making them better teachers. The remainder of this section reviews the evidence base motivating SOR exams, the literature on certification exams, teacher supply, and quality, and the Texas institutional context in which I test these predictions.

2.2 The Science of Reading and Teacher Preparation

The Science of Reading synthesizes decades of empirical research showing that reading acquisition requires explicit, systematic instruction in phonological awareness, phonics, fluency, vocabulary, and comprehension (Castles et al., 2018; Seidenberg, 2017). Meta-analyses consistently demonstrate the effectiveness of systematic phonics instruction for improving literacy outcomes (Ehri et al., 2001; McArthur et al., 2018). Despite this evidence, teacher preparation programs have historically emphasized approaches (e.g., balanced literacy and whole language) that downplay these methods, leaving classroom literacy instruction without an evidence-based foundation (Moats, 2020). Ellis et al. (2023) analyzed 702 elementary educator preparation programs across all fifty states and found that 30% provide no practice opportunities connected to the core components of reading. Since teacher knowledge of foundational literacy skills predicts student reading growth (Spear-Swerling & Brucker, 2004), and early, intensive, evidence-based instruction can bring most at-risk students to grade-level word-reading ability (Torgesen, 2004), inadequate preparation represents a primary barrier to equitable reading outcomes.

States have responded to this evidence aggressively. As of January 2026, 20 states require a standalone certification exam demonstrating minimum competency in the Science of Teaching Reading (National Council on Teacher Quality, 2024). In some cases, SOR reforms have coincided with improved student outcomes. For example, quasi-experimental evidence from Mississippi's comprehensive literacy reforms indicates gains of 0.23 standard deviations in reading (Spencer, 2024), and Novicoff and Dee (2025) find similarly positive effects of California's Early Literacy Support Block Grant. However, these initiatives bundle curriculum adoption, coaching, professional development, and certification requirements, making it difficult to isolate the contribution of licensure exams specifically. To date, no studies have directly examined whether Science of Reading certification exams improve student outcomes or how they shape the teacher workforce.

2.3 Teacher Certification Exams, Quality, and Supply

A large literature shows that certification requirements function as binding barriers to entry, shaping both the size and composition of the teacher workforce (Angrist & Guryan, 2008; Boyd et al., 2007; Deneault et al., 2025; Goldhaber, 2007). Angrist and Guryan (2008) demonstrate that more stringent testing requirements reduce teacher supply and raise teacher wages without measurable improvements in teacher quality. More recently, Deneault et al. (2025) use Texas certification exam data to document that narrowly failing a certification exam substantially delays entry into teaching and imposes significant earnings losses, with considerably larger effects for underrepresented minority (URM) candidates than for white candidates. These supply costs are especially salient at a moment of declining interest in the teaching profession (Bartanen & Kwok, 2023; Kraft & Lyon, 2024) and persistent gaps between the demographic composition of the teacher workforce and the students it serves.

Whether these barriers are justified depends on the exams' screening value, and here the evidence is less favorable. Teacher effectiveness has proven difficult to predict from anything observable at hire, with credentials and test scores exhibiting at most modest and inconsistent correlations and classroom experience among the few consistent predictors (Clotfelter et al., 2007; T. J. Kane et al., 2008; Rockoff et al., 2011). Some certification scores nonetheless carry predictive content (Cowan et al., 2023), but even where the score predicts, the passing *threshold*—the actual policy instrument—appears to screen poorly. Goldhaber (2007) finds a positive but small relationship between North Carolina licensure test scores and student achievement once teacher sorting is accounted for, and estimates that raising the state's cutoff to Connecticut's would exclude fewer than 0.5 percent of the workforce judged very ineffective while excluding 7 percent judged effective. Deneault et al. (2025) also speak to screening and exploit a reform that made Texas elementary certification exams more difficult. Here they find nearly identical value-added distributions among teachers who passed the easier and harder versions, so the harder exam reduced URM supply without improving quality. Orellana and Winters (2025) directly test whether licensure thresholds screen on quality using Connecticut data. Failing the Praxis II reduces the likelihood of certification by about 6.6 percentage points, with deterrent effects concentrated in shortage fields, and rather

than weeding out weaker candidates, failing disproportionately deters candidates at least as effective as those who passed. In sum, existing exams appear to carry limited screening information in their scores, and their thresholds function primarily as barriers.

Despite this evidence, most states continue to require and expand certification testing, for two understandable reasons. First, as preparation pathways grow more heterogeneous with the expansion of alternative certification (National Council on Teacher Quality, 2025), exams are one of the few mechanisms states retain to verify minimum competency. This concern is validated by evidence that teachers entering through relaxed pandemic-era emergency pathways were less effective than traditionally prepared peers (Backes et al., 2024; J. Kirksey & Gottlieb, 2023). Second, as states adopt new instructional approaches such as Science of Reading-based literacy instruction, new exams are created to verify these specific skills.

On the other hand, other states have considered this evidence and moved in the opposite direction. For example, New Jersey eliminated its Praxis Core requirement effective January 2025, citing shortages and financial barriers (NJ State Policy Lab, 2025). Similarly, several states have dropped or made optional the edTPA (Amin, 2022)—a performance-based, subject-specific assessment in which teacher candidates assemble a portfolio during their student teaching, including lesson plans, video of their own instruction, and written commentaries, scored externally to gauge classroom readiness.

Much of this policy debate treats the options as binary. States can maintain requirements that restrict supply and diversity, or they can weaken standards and risk admitting underprepared teachers. This study first asks whether a Science of Reading exam is worth its cost on those terms. It then examines an option the binary leaves out: helping candidates meet the standard rather than choosing whether to enforce it.

2.4 Texas' Science of Reading Exam

In June 2019, Texas passed House Bill 3, which mandated that new teacher candidates demonstrate proficiency in the Science of Reading through a certification exam. Beginning January 1, 2021, candidates pursuing certifications in early childhood through grade 6, core subjects grades 4–8, or English Language Arts grades 4–8 must pass the TExES Science of Reading exam (exam 293). The exam assesses knowledge of phonological awareness,

phonics, fluency, vocabulary, and reading comprehension instruction grounded in evidence-based practices. Example items are provided in Appendix B.

The exam consists of 90 multiple-choice questions scored on a scaled system with a passing score of 240. Candidates who fail may retake the exam after a 30-day waiting period, are limited to five attempts absent a waiver, and pay a fee of $116 per attempt, creating direct financial costs for candidates who must retake. The affected certifications account for 41.7% of newly issued teaching certificates in Texas in the pre-mandate period (Table 2), making the SOR exam a significant checkpoint in the teacher certification pipeline.

2.5 Texas Reading Academies at EPPs

To support preservice teachers' preparation in evidence-based early literacy instruction, the TEA authorized select EPPs to offer Texas Reading Academy coursework beginning in the 2024–25 academic year. These academies extend the TRA, a statewide professional development program rooted in the Science of Teaching Reading, to preservice preparation.

TRAs provide a structured, intensive sequence of coursework beyond standard EPP requirements. The curriculum consists of twelve modules covering the full scope of research-based literacy instruction, including oral language and vocabulary development, phonological and phonemic awareness, print concepts and alphabet knowledge, decoding and encoding, reading fluency, reading comprehension, and written composition. Instruction emphasizes explicit, systematic, and data-driven practices, with attention to culturally sustaining pedagogy, emergent bilingual learners, and special populations. Candidates complete online modules delivered through Canvas, participate in in-person instructional sessions, and demonstrate mastery through graded artifacts, assessments, and coaching, with a minimum mastery threshold of 80 percent across required assignments.

EPPs must apply through a TEA-administered authorization process requiring an Authorized Provider within the EPP, Cohort Leaders who pass a multi-stage screening, and required training and onboarding. Once approved, EPPs retain flexibility in integration: some offer academies as intensive standalone courses, while others embed the content across multiple semesters of coursework.

Importantly, participation is not universal. EPPs self-select into the application process,

not all applicants are approved, and adoption occurred at different points in time. Table A11 documents the authorized EPPs and each program's adoption year based on TEA administrative records.[1] This staggered and incomplete adoption generates variation in preservice teachers' access to academy coursework across programs and cohorts, which I exploit in the empirical analysis.

3 Data and Sample

This analysis uses administrative data from the University of Houston Education Research Center (ERC), which links records across multiple state agencies. The ERC integrates data from the TEA, the Texas Higher Education Coordinating Board (THECB), and other state entities through common identifiers, enabling longitudinal tracking of individuals across the K–12, postsecondary, and teacher workforce systems. I draw on four categories of data: teacher certification exams, teacher certifications, public school employment and student achievement, and postsecondary enrollment records.

3.1 Teacher Certification Exams and Certifications

The State Board for Educator Certification (SBEC) maintains two cumulative administrative files that make up the primary dataset used in this paper. The first is a record of all attempts at SBEC certification exams, covering every exam administered through March 2026. Each record includes the exam type, administration date, scaled score, pass/fail result, certification route (traditional, alternative, or certification by exam), and the EPP that approved the candidate to test. The second file contains all teaching certificates issued in Texas through early 2026. Each certification record includes the certification field, type, and program, the issue and effective dates, the EPP attended, and candidate gender and race/ethnicity. Both files contain unique identifiers that allow records to be linked across files and to other ERC data sources.

The exam file lacks demographics, so I construct race and ethnicity measures for exam takers by linking to two external sources: TEA's Public Education Information Management

[1] More information on participating EPPs can be found here.

System (PEIMS) student records, covering individuals who attended Texas public schools as K–12 students, and THECB postsecondary records, covering individuals who applied to or attended Texas public colleges and universities. Together, these linkages yield race and ethnicity for approximately 65 percent of first-attempt exam takers. The remainder likely attended K–12 school outside Texas and private or out-of-state postsecondary institutions. Analyses requiring demographics restrict to candidates with available covariates, and I note this restriction where it applies.

3.2 Public School Employment and Student Achievement

Employment data come from the PEIMS staff files, annual snapshots of all individuals employed in Texas public schools measured as of the last Friday in October. I observe employment status, district and campus assignments, role classifications, and experience at a single point each year, and I link these records to the exam and certification files to measure whether exam takers work as teachers in the October following their exam. For analyses linking teachers to students, I use TEA's class roster data, which connect teachers directly to the students in their classrooms. Student achievement is measured by State of Texas Assessments of Academic Readiness (STAAR) scores in reading and math, standardized to mean zero and standard deviation one within grade, year, and subject. Student covariates including race, gender, special education status, gifted status, and economic disadvantage, come from the PEIMS student files.

Value-added models that make use of this dataset use all regular STAAR administrations in grades 4 through 8 in reading and math from 2014 through 2024, excluding alternate and other specially scored assessments.[2] For candidates already teaching when they first sit for the SOR exam, this includes years before the exam. Each teacher's value-added is therefore a single estimate pooled across all observed years.

[2]Grade 3 students are excluded because the models control for prior-year achievement and STAAR testing begins in grade 3, so no lagged score exists for them.

3.3 First-Attempt Certification Exam Takers

The first analytic sample consists of all individuals who sit for the TExES SOR exam for the first time with a valid scaled score, restricted to candidates whose exam date falls at least one year before the end of the employment data.[3] This ensures a minimum of one year of follow-up for certification and employment outcomes. The resulting sample includes 21,692 first-attempt SOR exam takers from January 1, 2021 through October 23, 2023.

Table 1 presents summary statistics overall and by first-attempt pass status. Approximately 80 percent of candidates pass on their first attempt, with a mean scaled score of 251.6 against a passing score of 240. Candidates who fail are disproportionately on the alternative certification route (52.7 percent, versus 36.3 percent of those who pass), more likely to be non-white (53.0 versus 40.2 percent among the subsample with demographics), and slightly less likely to be female (89.0 versus 93.0 percent).

A notable feature of the data is that approximately 31 percent of first-attempt SOR exam takers are already employed as teachers in Texas public schools before their exam date, at nearly identical rates among those who pass (31.6 percent) and fail (31.1 percent). This reflects the structure of the Texas teacher labor market, in which individuals can teach on interim credentials, emergency permits, or without formal certification in districts of innovation.[4] Accordingly, 58.8 percent of candidates earn a Standard certificate within one year while 70.8 percent are employed as teachers the following October.[5] This is consistent with a context where many teachers enter classrooms before completing all certification requirements.

3.4 Newly Issued Certifications

The second analytic sample consists of all newly issued standalone teaching certificates in Texas from 2016 through 2025, where each observation is the first time an individual receives

[3]The scaled-score restriction also excludes the exam's initial piloting period, during which SBEC reported only pass/fail results, so the effective sample begins in September 2021 rather than January 2021.

[4]These are districts who are allowed by TEA to opt out of regulations on inputs like teacher certification and class sizes. See Anglin (2024) for more information.

[5]A Standard certificate is the full five-year renewable teaching credential. Candidates may teach before earning one on an intern, probationary, or emergency certificate, or without any certificate in districts of innovation.

a particular certification (i.e., individuals who earn multiple standalone certifications may appear multiple times in the dataset).[6] Here, I observe the certification field, type, issue date, EPP, and candidate demographics for all 342,816 records. I classify each certificate as SOR-required or comparison following the Texas Education Agency Required Test Chart. Table A2 lists the full classification.[7]

Table 2 presents summary statistics for this sample, split by whether the certification was issued before or after the SOR mandate's June 2019 announcement. The two periods are broadly similar in route composition, though out-of-state certifications rise modestly, from 9.9 to 13.6 percent. Descriptively, the share of newly issued certifications in SOR-required fields fell from 41.7 percent pre-announcement to 36.0 percent afterward. The demographic composition also shifted, with the White share declining from 59.2 to 55.2 percent and the Hispanic share rising from 25.1 to 27.6 percent.

To estimate how the SOR requirement affected the flow of new certifications, I collapse this sample to counts by educator preparation program (EPP) $\times$ program-type (e.g., traditional, alternative, etc.) $\times$ certification-field $\times$ half-year (i.e., January to June and July to December).[8] This collapse spans 200 EPP-program-type units, 3 SOR-required and 16 comparison certification fields, and 2016 January to June (H1) through 2025 July to December (H2), yielding 9,585 SOR-required and 51,120 comparison cells. Table 3 describes the resulting panel.[9]

[6] I begin the sample in 2016 to avoid the January 2015 replacement of the Generalist EC-6 (TExES 191) and Generalist 4–8 (TExES 111) exams with the Core Subjects EC-6 (TExES 291) and 4–8 (TExES 211) exams.

[7] A certification is coded SOR-required if the TExES Science of Teaching Reading exam (293) is among its required content-pedagogy tests. Non-teaching credentials and supplemental certificates that attach to a base certificate rather than certifying a teaching field independently are excluded from the sample. For more information, see the notes to Table A2.

[8] Some EPPs offer multiple program types. For example, several universities that offer traditional teacher education programs to undergraduates also operate alternative certification programs.

[9] I code a cell as zero if the EPP-program-type issued no certifications in that field during that half-year but was otherwise active, defined as having issued a certification in *any* field during that half-year. Periods before an EPP-program-type's first observed activity, or after its last, are excluded from the panel rather than coded as zero. This distinguishes a program that was operating but chose not to issue a given certification from one that had not yet begun, or had ceased, issuing certifications altogether. Under this rule, 51.2 percent of SOR-required cells and 69.9 percent of comparison cells are zero, reflecting the narrowness of many comparison fields (e.g., Foreign Language, Computer Science) relative to the broader SOR-required fields.

3.5 Texas Reading Academy Data

For the analysis of Texas Reading Academies, I obtained the set of EPPs authorized to offer academy coursework, and each program's adoption year, from TEA administrative records (Table A11). I match this information to EPP identifiers in the exam and certification files to classify candidates by whether their EPP offered a TRA at the time of their first exam attempt.

Table 4 compares pre-period characteristics of the 10 EPPs that adopted academies in 2024–25 and the 98 EPPs that never adopted, measured over the 2021–22 through 2023–24 academic years. EPPs adopting in 2025–26 are excluded from both groups. Most importantly, the two groups perform similarly on both the SOR and Pedagogy and Professional Responsibilities (PPR) exams in the pre-period. Pass rates and scaled scores are statistically indistinguishable for both exams. The groups are also balanced on certification production, demographics, and the share of certifications in SOR-required fields. The most notable compositional difference is that academy EPPs are more likely to be traditional university-based programs (73.3 versus 55.8 percent).

4 Empirical Strategy

4.1 Barrier: The Cost of the Requirement

4.1.1 Individual-Level Cost of Failing

Candidates scoring 240 or above pass the SOR exam and those below fail. Assuming candidates just above and just below this cutoff are comparable in all respects other than pass status, comparing their outcomes identifies the causal effect of passing the exam for candidates at the margin.

Using a regression discontinuity at the passing threshold, I examine four outcomes that trace the consequences of failing from immediate response to career trajectory: whether candidates retake the SOR exam within one year, whether they earn a Standard teaching certificate within six months and within one year of their exam date, and whether they work

as a teacher in a Texas public school the following October.[10]

Formally, I estimate

$$Y_i = \alpha + \beta \cdot \mathbb{1}(\text{Score}_i \geq 240) + f(\text{Score}_i - 240) + \varepsilon_i, \tag{1}$$

where $f(\cdot)$ is implemented as a local linear regression with a triangular kernel and the data-driven bandwidth of Calonico et al. (2014). Since employment is observed only annually while exam dates vary throughout the year, exam timing mechanically determines how long candidates have to secure a position before the October snapshot. For consistency, all specifications therefore include month-of-exam fixed effects. I also assess robustness to alternative bandwidth selectors. Since certification routes differ in candidates' institutional position at the time of the exam, with some finishing university-based programs and others already teaching on interim credentials, I estimate the model separately by route.

Identification requires that candidates just above and below the cutoff are comparable in all respects other than pass status. The main threat to this is manipulation. If candidates could precisely control their scores, those who landed just above the threshold might differ systematically from those just below. A formal density test (Cattaneo et al., 2018) narrowly fails to reject the null of no manipulation ($p = 0.051$), and the score density rises smoothly through the threshold with no visible bunching on either side (Figure A2). A p-value this close to conventional thresholds warrants scrutiny, so it is worth stating why manipulation is implausible in this setting. Scaled scores are computed from raw scores through a conversion that candidates do not know in advance, candidates cannot observe their standing relative to the threshold while testing, and no mechanism exists to revise a score once submitted. A candidate who wanted to land just above 240 would have no way to do so. The observable evidence is consistent with this. Pre-test covariates are balanced across the threshold (Table A1). Candidates just above and just below do not differ in certification route, demographics, or prior teaching status. Manipulation is therefore extremely unlikely in this setting.

[10] Teachers are observed in Texas public schools once annually. For example, for a candidate who takes the SOR exam between October 1, 2023, and September 30, 2024, employment is measured as of the October 2024 staffing snapshot for the 2024–25 school year.

4.1.2 Market-Level Effects on the Flow of New Certifications

The RD identifies the cost of failing for candidates near the threshold, but not the margins where the mandate's barrier may matter most. Candidates who score well below the cutoff face more uncertain paths to passing, and the exam's direct and indirect costs (e.g., fees, study time, travel to a testing center) may deter prospective candidates from entering the pipeline at all. To examine this margin, I compare the number of newly issued standalone teaching certifications in SOR-required fields to the number in comparison fields, using the sample described in Section 3.4.

Using counts of newly issued certifications at the educator preparation program (EPP) by program type (e.g., Traditional, Alternative) by certification field (e.g., EC–6 Core Subjects) by half-year level, I estimate event study regressions via Poisson pseudo-maximum likelihood:

$$\mathbb{E}\left[Y_{ekct}\right] = \exp\left(\sum_{t \neq 2019\text{H1}} \beta_t \cdot \mathbb{1}(t) \times \text{SOR}_c + \gamma_{ek} + \delta_c + \lambda_t\right), \tag{2}$$

where Y_{ekct} is the number of new certifications issued by EPP e through program type k in certification field c during half-year t, and SOR_c indicates that field c began requiring the SOR exam in January 2021. The specification absorbs EPP-by-program-type (γ_{ek}), certification field (δ_c), and half-year (λ_t) fixed effects, so identification comes from within-EPP-program-type comparisons of how SOR-required and comparison fields evolve over time. The exponentiated β_t give the rate ratio of certifications in SOR-required relative to comparison fields in each half-year, normalized to 2019H1, the last period before House Bill 3 was enacted in June 2019. PPML accommodates the count nature of the outcome and allows estimation with cells containing zero certifications (Santos Silva & Tenreyro, 2006).[11] Standard errors are clustered at the certification field level, the level at which the SOR requirement is assigned.

[11] PPML is the preferred estimator here for two reasons. First, 51.2 percent of SOR-required cells and 69.9 percent of comparison cells are zero, so log-linear models are undefined without an arbitrary transformation of the outcome. Second, exponentiated PPML coefficients are rate ratios and so are invariant to program scale, which matters because EPPs range from those issuing a handful of certifications a year to those issuing several hundred. PPML is also consistent under any conditional mean of the form in Equation 2, this means the outcome variable need not be Poisson distributed, and its fixed effects are free of the incidental parameters problem (Wooldridge, 1999).

I summarize the event study with three pooled specifications, each estimated overall and separately by certification route. The first replaces the half-year interactions with a single indicator for all periods after 2019H1. This captures the average post-announcement effect, including any anticipatory response. Once candidates learned of the mandate in June 2019 they had an immediate incentive to certify before the exam became a requirement, and restricting attention to post-2021 changes would understate the requirement's full effect on the flow of entrants. The second is a donut specification that drops 2019H2 through 2021H2 from the estimation sample and defines the post period as 2022H1 onward, removing the anticipatory window and its immediate aftermath rather than folding it into the post indicator. I interpret this as the effect once the market has settled into a new steady state. The third addresses the possibility that the two groups were already diverging before the announcement. It adds the term $\text{SOR}_c \times t$ for $t \leq$ 2019H1, allowing the differential between SOR-required and comparison fields to follow its own linear trend through the pre-announcement period. The counterfactual for the post period is then the level that fitted trend reaches in 2019H1 rather than the average pre-announcement differential, so the pooled coefficient is not driven by a gap that was already moving before House Bill 3. I report all three throughout.

With 19 certification field clusters, 3 of which are SOR-required, the asymptotic justification for cluster-robust standard errors is weak, and with few treated clusters conventional standard errors can understate the sampling variability of the treatment coefficient (Cameron & Miller, 2015). I therefore assess inference on each pooled specification via randomization inference, following MacKinnon and Webb (2020). I re-estimate the specification 2,000 times, each time randomly assigning SOR-required status to 3 of the 19 certification fields while holding the number of treated fields fixed, and record the t-statistic on the coefficient of interest. The reported p-value is the share of placebo draws whose t-statistic equals or exceeds in magnitude the t-statistic from the true assignment.[12][13] I report the randomization inference p-value alongside, rather than in place of, the clustered p-value, and discuss any

[12] The wild cluster bootstrap is the more common alternative in linear settings with few clusters, but has no off-the-shelf implementation for fixed-effects PPML. Randomization inference does not require asymptotics to kick in, so the results remain valid however few certification fields the sample contains.

[13] As a further check, I re-estimate the event study and all three pooled specifications by ordinary least squares and wild cluster bootstrap with Webb weights, which are feasible for linear models. Despite the model parsimony, this specification is not preferred because its coefficients are in number of certificates and difficult to interpret across EPPs of different sizes.

divergence between them in the results.

Interpreting the post-mandate change as the effect of the requirement rests on the assumption that, absent the mandate, certifications in SOR-required fields would have evolved in parallel to those in comparison fields. This assumption is untestable, but three pieces of evidence bear on it. Figure 3a plots raw certification counts for the two groups over the full panel. The event study provides pre-announcement coefficients, for which I report a joint test of the null that all are zero. The third pooled specification relaxes parallel trends directly by allowing the two groups to differ in level and in pre-announcement slope. None of these rules out a smooth pre-existing divergence that continues into the post period, so in Table A6 I report how large a violation of parallel trends would have to be to overturn the pooled estimate, following Rambachan and Roth (2023).

Two confounders deserve particular attention. The first is the COVID-19 pandemic, which arrived several months before the mandate took effect. If the pandemic differentially affected demand for or supply of teachers across certification fields, by making elementary teaching relatively less attractive or shifting hiring toward fields such as special education, the estimates would partly reflect those changes rather than the SOR requirement. The second is that the mandate coincided with a redesign of the content exams in the affected fields. When the standalone SOR exam was introduced, reading pedagogy content was removed from the EC–6 and ELAR 4–8 content exams and revised versions were launched. The estimates therefore capture the combined effect of the new testing regime rather than the SOR requirement in isolation.

4.2 Screen: Does the Exam Identify Effective Teachers?

The welfare implications of the exam's supply effects depend on whether it screens for knowledge that matters for teaching effectiveness. If the exam identifies teachers who are more effective in the classroom, the costs documented above may be offset by gains in teacher quality among those who enter the profession. This section evaluates that possibility on two margins, whether exam scores predict teacher effectiveness and whether the candidates the passing threshold excludes are less effective than the ones it admits.

4.2.1 Predictive Validity of Certification Exam Scores

One way a certification exam may function as a screen is by providing information about teacher effectiveness before hire. A substantial literature has searched for such signals and largely come up short, though certification exam scores are among the few characteristics that consistently—if modestly—correlate with future effectiveness (Clotfelter et al., 2007; T. J. Kane et al., 2008; Rockoff et al., 2011). This section asks whether the SOR exam is such a signal, and whether it is a more informative one than a generic pedagogy exam like the PPR.

To assess whether certification exams predict teacher effectiveness, I estimate student-level value-added models of the form

$$A_{ijt} = \gamma \cdot \widetilde{S}_j + X'_{ijt}\beta + \bar{X}'_{cjt}\delta + \mu_t + \lambda_g + \varepsilon_{ijt}, \tag{3}$$

where A_{ijt} is the standardized STAAR score of student i taught by teacher j in year t and $\widetilde{S}_j$ is teacher j's first-attempt scaled score on the relevant certification exam, standardized within test takers. The vector X_{ijt} holds student-level controls: lagged STAAR exam scores in reading and math, entered with quadratic and cubic terms, and indicators for sex, race/ethnicity, limited English proficiency, special education status, gifted status, and economic disadvantage. The vector $\bar{X}_{cjt}$ holds classroom-level averages of these same covariates, absorbing the peer composition a student faces and the composition of the school as a whole. Year (μ_t) and grade (λ_g) fixed effects absorb statewide shocks common to a testing year and level differences across grades, and ε_{ijt} is the student-level error, clustered at the teacher level. Additional specifications add district and campus fixed effects, so that γ is identified from comparisons of teachers within the same district or the same school rather than across them. I estimate Equation 3 separately for the SOR and Pedagogy and Professional Responsibilities (PPR) exams and interpret γ as the strength of a signal of teaching quality observable before hire.

To test directly whether the SOR exam carries information beyond the PPR exam, I also estimate horse-race specifications that replace $\widetilde{S}_j$ with both scores entered simultaneously, so that each coefficient reflects the association with student achievement net of the other.

A positive and statistically meaningful SOR coefficient conditional on the PPR score would indicate the SOR exam captures information not already contained in the general pedagogy exam.

4.2.2 Which Candidates Does the Threshold Exclude?

The predictive validity of the score does not tell a state what it gains or loses by drawing a line through it. That depends on whether the candidates the threshold excludes would have been less effective teachers than the ones it admits. To examine this, I estimate the following regression via OLS:

$$VA_j = \alpha + \beta \cdot \text{Fail}_j + \varepsilon_j, \qquad \underline{S} \leq S_j \leq 250, \tag{4}$$

where VA_j is teacher j's estimated value-added in reading, Fail_j indicates a first-attempt SOR score below 240, S_j is the first-attempt score, and ε_j is the error term, against which I report heteroskedasticity-robust standard errors.[14]

$\underline{S} \leq S_j \leq 250$ indicates that the sample is restricted on both sides of the cutoff. Passers are limited to those who scored no more than ten points above the threshold (i.e., ∼0.58 standard deviations), so that the comparison group consists of candidates who cleared the bar by a fairly narrow margin. Failers are limited to those scoring at or above a floor $\underline{S}$, which I lower in ten-point steps from ten points below the cutoff (about the 5th percentile of first-attempt scores) to forty points below it (about the 0.5th percentile). The narrowest window compares candidates on either side of the cutoff who are close to one another in score. Each wider window adds failers further below the cutoff, so the sequence of estimates traces how the gap changes as the failers included come to resemble the candidates a progressively lower threshold would admit.

[14]Value-added is estimated from Equation 3 with $\widetilde{S}_j$ omitted and a teacher fixed effect included in its place, so that VA_j is the unshrunk teacher effect. I use fixed rather than random effects for two reasons. First, value-added enters Equation 4 as the outcome rather than a regressor, so estimation error in VA_j widens the standard error on β without biasing it, and the precision gain from shrinkage is not needed for consistency. Second, shrinkage pulls the estimates of teachers linked to fewer students toward the sample mean. If first-attempt failers enter teaching later than passers and are therefore linked to fewer students over the sample period, a shrunken estimate would compress failers' value-added toward the mean more than passers', biasing β toward zero.

The estimate of β is biased if the failers who appear in the value-added sample are not representative of failers generally. There is reason to expect they are not, since failing itself can keep candidates out of the classroom. I describe this selection in two ways. First, I re-estimate Equation 4 in the same four windows with the outcome replaced by an indicator for being observed with a reading value-added estimate. This measures how strongly failing predicts dropping out of the sample. Second, among first-attempt failers, I estimate

$$X_j = \alpha + \delta \cdot \text{Observed}_j + \gamma \cdot S_j + \theta_{r(j)} + \varepsilon_j, \tag{5}$$

where X_j is one of four pre-exam characteristics, an indicator for being female, an indicator for being white, an indicator for family income below \$60,000, and SAT score; Observed_j indicates having a value-added estimate; and $\theta_{r(j)}$ is a fixed effect for candidate j's certification route, so the comparison is within route. I estimate Equation 5 with and without a candidate's first-attempt score as a control. The coefficient δ measures how failers who appear in the value-added sample differ from failers who do not, and its sign indicates the direction in which selection moves β. For example, if failers who go on to teach have higher SAT scores than failers who do not, the observed failers are stronger on academic preparation than failers as a whole. Assuming candidates with higher SAT scores have higher value-added, β would then understate the gap between narrow passers and the full population of failers.

4.3 The Effect of Texas Reading Academies

The final analysis tests whether structured coursework can help candidates clear the bar without lowering it. To do this, I compare exam outcomes at EPPs that did and did not offer TRAs, before and after implementation, using a difference-in-differences design. The data are first-attempt exam records, and the outcomes are an indicator for passing the SOR exam on the first attempt and the first-attempt scaled score, standardized within year.[15]

[15]Because additional EPPs adopted academies in 2025–26, and to avoid comparing already-treated to not-yet-treated programs under staggered adoption (Goodman-Bacon, 2021), I separately analyze the 2024–25 and 2025–26 cohorts against a never treated comparison group of EPPs.

The event study specification is

$$Y_{iet} = \alpha_e + \delta_t + \sum_{k \neq 2023} \beta_k \cdot \text{Academy}_e \times \mathbb{1}(t = k) + \varepsilon_{iet}, \tag{6}$$

where Y_{iet} is the first-attempt outcome of candidate i at EPP e in academic year t, α_e are EPP fixed effects, δ_t are year fixed effects, and standard errors are clustered at the EPP level, the level at which academies are adopted. Because the regression is estimated on candidates, EPPs contribute in proportion to their number of test takers. The β_k measure academy–non-academy differences in year k relative to the year before implementation. A companion difference-in-differences model replaces the interaction terms with a single $\text{Academy}_e \times \text{Post}_t$ indicator.

Identification again rests on parallel trends. In this case, this means that absent the academies, outcomes at adopting and non-adopting EPPs would have evolved similarly. Two features of the design support the plausibility of this assumption for the 2024–25 cohort. First, the pre-period event study coefficients provide a test of differential trends before EPPs select into the academies. Second, adopting and non-adopting EPPs are balanced on pre-period pass rates and scores for both the SOR and PPR exams (Table 4). I focus the main analysis on this cohort for two further reasons. The 2025–26 cohort's pre-period event study coefficients reject the joint test of parallel trends, so a difference-in-differences comparison for this cohort is not well identified, and its post-adoption window covers only a partial school year, providing limited outcome data even setting the pre-trend concern aside. I report the 2025–26 cohort's results separately in Appendix Table A12 for completeness rather than as a corroborating estimate. Since I exclude one of the two academy cohorts from the main analysis, I interpret these results as preliminary rather than settled evidence on whether structured preparation affects exam performance.

5 Results

The SOR exam first-attempt pass rate hovers around 80 percent over the sample period, which is roughly comparable to the PPR exam and well above the EC-6 Content exam

(Figure A1). The level is high enough that most candidates eventually pass. It is also low enough that one in five fails on the first attempt and faces exam fees, preparation time, and delayed certification. The analyses below test the extent to which the exam is a barrier at the individual level (Section 5.1) and the market level (Section 5.2), whether it screens on teacher effectiveness (Section 5.3), and whether preparation can mitigate its supply cost (Section 5.4).

5.1 The Individual Cost of Failing

The most immediate consequence of failing is retaking. Table 5 reports estimates of Equation 1 for each outcome, and passing reduces the probability of retaking the exam within a year by 77.9 percentage points. The majority of candidates who narrowly fail make at least one additional attempt. The more substantive question is whether failure delays or prevents certification and employment. Passing increases the probability of earning a teaching certificate within six months by 9.2 percentage points. By twelve months the estimate attenuates to 4.9 percentage points and loses statistical significance, which indicates that most candidates who narrowly fail eventually retake, pass, and certify. The friction is not fully absorbed by the certification process, however. Passing increases the probability of working as a teacher the following October by 6.5 percentage points. This effect is not purely mechanical. Individuals in Texas can and do teach without a certificate, and roughly a third of test takers are already in classrooms before their exam (Table 1), so failing does not by itself bar a candidate from employment. Where it does operate as a bar, through program rules that condition placement on passing, the estimates by route below show it. Figure 1 shows these discontinuities graphically.

These results are robust. Point estimates are nearly identical with and without month-of-exam fixed effects (Table A3) and stable across alternative data-driven bandwidth selectors (Table A4). Some estimates lose statistical significance without controls or under narrower bandwidths, but in each case the point estimates barely move reflecting that these decisions result in differences in precision, not in the qualitative result.

The overall results conceal a sharp divide by certification route (Table 6). Among traditional candidates, who are typically finishing university-based programs and not yet em-

ployed as teachers, the exam comes at the end of the pipeline. Those who narrowly fail retake at very high rates (92.4 percentage points), and passing at the margin raises six-month certification by 16.0 percentage points. These results are consistent with candidates sitting for the exam on the cusp of completing all other requirements. By twelve months the gap in the share of certified candidates above and below the cutoff closes entirely, with a small, wrong-signed, insignificant point estimate (−1.7 percentage points), and there is no effect on employment in the subsequent October. For a traditional candidate who just fails, the exam delays a credential they were about to receive without derailing the career behind it.

The pattern differs for alternative route candidates. Even among alternative candidates who pass, fewer than a quarter certify within a year (Figure 2, panel (c)), far below every other route. The certification effect of passing is small at six months (2.5 percentage points) but grows to 10.6 percentage points at one year ($p < 0.001$), and narrowly failing reduces the probability of teaching the following October by 12.0 percentage points. This pattern is consistent with where the exam comes in the alternative route's sequence. Many alternative programs require candidates to pass the relevant certification exams before they can be placed on an intern certificate, so passing the exam is typically one of the first steps toward the classroom rather than one of the last, and failing it blocks entry to that pathway altogether.

The out-of-state and certification-by-exam routes fit the same interpretation. Failing lowers six-month certification by 21.8 and 16.0 percentage points, respectively, but the gap shrinks by one year and there is no meaningful employment effect, though both samples are small enough that the estimates are imprecise. This pattern fits candidates who typically hold a teaching certificate already and are sitting for the SOR exam to add a field rather than to enter the profession. Put differently, for these candidates, the exam comes after most of the work of becoming a teacher is already done.

These results show that the cost of narrowly failing depends on where the exam falls in a candidate's route to certification. It comes at the end of a sequence for traditional candidates, at the entry point for alternative candidates, and after certification is already secured for out-of-state and by-exam candidates. But estimates at the passing margin capture only candidates who show up to test and just miss. If the exam also deters people from attempting

at all, the fuller effect on supply appears not at the cutoff but in the flow of new certifications, the subject of the next section.

5.2 The Market-Level Cost

Panel (a) of Figure 3 shows the market-level pattern of newly issued standalone certifications in SOR-required and non-SOR fields by half-year. Both series show a pronounced seasonal sawtooth, peaking each January–June as candidates completing traditional programs certify ahead of spring graduation, and both drift downward from 2016 through 2019, consistent with the broader national decline in entry into teaching over this period (Bartanen & Kwok, 2023; Kraft & Lyon, 2024). The decline is somewhat steeper in SOR-required fields, a point I return to below. The COVID-19 pandemic mutes both series' January–June 2020 peak but disrupts neither series relative to the other. Against this backdrop, the divergence around the mandate is clear. Certifications in fields that would require the SOR exam surge in the second half of 2020, breaking the usual seasonal trough, as candidates rush to certify before the requirement took effect. The series then drops sharply once the exam became required in January 2021 and settles onto a lower path. The gap between the two series widens at the mandate and has not closed through 2025.

Panel (b) formalizes this pattern by plotting the coefficients estimated from Equation 2. Each point represents the estimated rate ratio of certifications in SOR-required relative to non-SOR fields, normalized to the first half of 2019. Pre-announcement rate ratios are statistically indistinguishable from one, and a joint test fails to reject the null that the pre-period coefficients are jointly zero ($p = 0.638$; Table 7). They do, however, decline gradually from about 1.09 in 2016 to 1.00 in the reference period, so SOR-required fields were losing ground to comparison fields slowly before the announcement. The rate ratio rises to 1.36 in the second half of 2020 as candidates accelerated certification ahead of the mandate, then falls to 0.67 in the first half of 2021 as that pipeline exhausted. From 2022 onward the rate ratio declines from 0.77 to 0.57, so the gap widens rather than closes over the four years following implementation. Taken at face value, an EPP that would have issued one hundred certifications in SOR-required fields, given its own history and the statewide trend in comparison fields, issues seventy-seven in 2022 and fifty-seven by 2025. Part of that decline

continues the pre-announcement drift, which the trend-adjusted estimate below nets out.

Table 8 condenses these dynamics into single estimates. Including the anticipatory period, certifications in SOR-required fields fell by 21.5 percent ($0.785 - 1 = -0.215$) relative to non-SOR fields within EPPs. The donut specification, which excludes half-years 2019H2–2021H2 to remove the anticipatory surge and its immediate aftermath, implies a larger decline of 31.4 percent ($0.686 - 1 = -0.314$). The trend-adjusted specification, which allows the pre-announcement differential to continue on its linear path, implies a decline of 17.8 percent ($0.822 - 1 = -0.178$), smaller than the other two because part of the raw gap reflects the slow pre-announcement drift documented above. I treat this as the preferred estimate of the mandate's effect, since it is the one that does not attribute the pre-existing divergence to the requirement. All three estimates are significant at the 5 percent level under clustered standard errors ($p = 0.016$, 0.004, and 0.002). Under randomization inference the evidence is weaker, with p-values of 0.128, 0.107, and 0.096 for the full, donut, and trend-adjusted specifications. With only three of nineteen certification fields treated, randomization inference is the more conservative procedure, and I read the market-level decline as a robust point estimate whose statistical significance depends on the inference method. Estimating the same specifications by ordinary least squares yields the same pattern (Table A5).

Sensitivity to violations of parallel trends points the same way. Table A6 shows that, allowing the pre-period trend to continue linearly into the post period, the estimated decline over 2019H2–2025H2 is significant at the 10 percent but not the 5 percent level, while the estimated decline restricted to the post-implementation period, 2022H1–2025H2, is significant at the 5 percent level. Both estimates are sensitive to allowing the pre-period trend to bend beyond a small threshold. The weaker result for the full post-announcement period is consistent with its inclusion of the second-half-2020 anticipatory surge.

Columns (2)–(5) of Table 8 report the same three specifications separately by certification route, and Figure 4 plots the corresponding event studies. The contraction is concentrated outside the traditional, university-based route. Under the trend-adjusted specification, certifications fell by 17 percent in the alternative route, 24 percent out of state, and 40 percent in the certification-by-exam route, while the traditional route shows a decline of under 3 percent that is statistically indistinguishable from zero under either inference method. The

full specification gives the same ordering; in the donut, the alternative and out-of-state estimates swap places. This is consistent with the RD finding that a narrow failure is only a temporary setback for candidates embedded in university-based preparation, and a larger obstacle for those entering through other routes.

The route-specific estimates warrant caution. The joint pre-period test rejects for the traditional, alternative, and by-exam routes (Table 7), but the pre-trend problems differ by route. For traditional and out-of-state candidates, the pre-period coefficients are tightly estimated and hover near one without a discernible direction, so the traditional rejection reflects precision rather than an economically meaningful trend.[16] For alternative and by-exam candidates, the pre-period coefficients decline steadily in the run-up to the announcement, running in the same direction as the post-mandate effect. The drift is most severe for by-exam, whose rate ratio falls from 1.31 to 1.10 over the six pre-periods. The trend-adjusted specification is designed to absorb a linear version of this drift, and its by-exam and out-of-state estimates remain significant under randomization inference ($p = 0.043$ and 0.059), while the alternative-route estimate does not ($p = 0.114$). Because a pre-trend that is not linear would survive this adjustment, I do not treat the alternative and by-exam estimates as cleanly identified, and I read the route-level results as a rough decomposition of where the aggregate decline is concentrated rather than as four separately identified effects. The pooled estimate, whose own pre-trend is modest and whose joint test does not reject, remains the primary market-level finding.

One possible explanation for the decline is substitution, with candidates redirecting toward non-SOR fields rather than exiting or delaying. I find no evidence for this. At the passing margin, the regression discontinuity in Table A8 splits the certification outcome by field. Failing the exam sharply reduces the probability of certifying in an SOR-required field, by 10.6 percentage points within six months and 8.5 within a year, while the effect on certifying in a non-SOR field is small and statistically indistinguishable from zero at both horizons. Candidates who narrowly fail do not pivot into unaffected fields. At the market level, certifications in comparison fields would have risen to absorb candidates leaving the

[16] The joint test does not reject for out-of-state ($p = 0.290$), and its pre-period coefficients are similarly flat.

affected fields, and the event study in Figure 3 shows no such offsetting increase. This is consistent with the mandate's effect being a contraction in the affected fields and not a reshuffling across them.

5.3 Does the Exam Screen?

The supply costs documented above are only worth bearing if the exam measures knowledge that matters for student learning. Table 9 reports estimates of Equation 3, relating student achievement on the STAAR reading (Panel A) and math (Panel B) exams to their teachers' SOR and PPR certification exam scores.

The reading results provide the most direct test of domain-specific screening. A one standard deviation increase in a teacher's SOR score is associated with a 0.011 standard deviation increase in student reading achievement in the baseline specification (column 1). The coefficient grows to 0.013 when comparisons are restricted to teachers within the same district (column 2) and school (column 3), which indicates the relationship is not driven by higher-scoring teachers sorting into higher-performing schools. All three estimates are statistically significant. To put the magnitude in context, the standard deviation of teacher value-added in reading in these data is 0.18 student standard deviations, so a one standard deviation increase in SOR score corresponds to about 0.07 standard deviations in teacher effectiveness. This is modest in absolute terms but comparable to the strongest associations between licensure test scores and student achievement reported elsewhere (Clotfelter et al., 2007; Cowan et al., 2023; Goldhaber, 2007). The corresponding PPR estimates are substantially smaller. They are 0.003 and insignificant at baseline and reach only 0.005 in the fixed-effects specifications (columns 4–6).

Entering both scores together sharpens the comparison. Columns (7)–(9) include SOR and PPR scores in the same model, so each coefficient reflects the association net of the other. In both reading and math, the SOR coefficient is essentially unchanged, while the PPR coefficient is small and insignificant throughout. Whatever predictive content the PPR score carries on its own is absorbed by the SOR score, and the reverse is not true.

Whether the SOR score measures knowledge specific to literacy instruction is harder to say. Panel B shows that SOR scores predict student math achievement at least as strongly

as reading.[17] This might be explained in two ways. First, literacy is foundational to learning in every subject, so a teacher who is better at teaching reading may raise math achievement through students' ability to read and understand math problems. Second, the exam may capture a general dimension of instructional skill that the PPR exam measures poorly or misses, so that its score predicts effectiveness in any subject rather than in reading specifically. The reading estimates may also understate what the exam measures, since value-added is observed only in grades 4 through 8 and the knowledge the exam tests is most relevant to teaching emergent readers in earlier grades. What the results do show is that the SOR score is a better pre-hire indicator of teacher effectiveness than the PPR score, and that the PPR score adds nothing to it.

While exam scores predict teacher effectiveness, the more policy-relevant question is whether the candidates the threshold excludes would have taught reading less well than the ones it admits. Panel A of Table 10 reports estimates of Equation 4, comparing first-attempt failers who went on to teach with candidates who passed within ten points of the cutoff and widening the set of failers included in each successive column. In every window, failers have reading value-added 0.016 to 0.021 standard deviations below narrow passers, about 0.09 to 0.12 standard deviations of the distribution of teacher effectiveness in reading. The gap is not statistically distinguishable from zero when the comparison is limited to failers within ten or twenty points of the cutoff, but becomes so once failers thirty or more points below it are included (columns 3 and 4, $p < 0.05$). Because most failers score within ten points of the cutoff, widening the window this far adds relatively few candidates, about 3.6 percent of all first-attempt takers, and the widest window still excludes only the 0.6 percent who score below 200.

Two readings of this pattern are possible. If the point estimate is stable and only precision improves as the window widens, then failers just below the cutoff are about as far behind narrow passers as failers generally, and even a small reduction in the threshold would admit less effective teachers. If instead the gap grows with distance from the cutoff, as the larger estimate in the widest window suggests, then the candidates a much lower threshold would

[17] The standard deviation of teacher value-added is 0.24 student standard deviations in mathematics compared with 0.18 in reading, so there is less variation in teacher effectiveness for a reading-focused exam to detect.

admit are weaker still. The data cannot separate these, but under either reading first-attempt failers are, on average, less effective at raising reading achievement than candidates who narrowly passed.

Columns (5)–(8) quantify the selection this comparison rests on. Failing on the first attempt lowers the probability of being observed with a reading value-added estimate by 1.5 percentage points in the narrowest window and by 3.8 percentage points in the widest, with the gradient significant from the second window onward. Among candidates observed teaching the following October, the same regressions produce estimates near zero (Table A9), so failers drop out of the value-added sample by not reaching a classroom rather than by being assigned to untested grades and subjects once there. Panel B reports estimates of Equation 5 and describes which failers do reach one. Relative to failers who are never observed with value-added, observed failers are 3.3 percentage points less likely to be female, 6.4 percentage points less likely to come from a family earning under $60,000, and score about 25 points higher on the SAT, with no difference in the share who are white. Controlling for first-attempt score leaves these estimates essentially unchanged, so the pattern is not a byproduct of higher-scoring failers being more likely to teach. The comparison group of passers displays selection of similar sign and size (Table A10), so differential selection between the two sides of the comparison is modest and unlikely to account for the gap in Panel A.

The direction of this selection matters for how to read the estimate. A lower threshold would admit failers who currently do not teach, and Panel B indicates those candidates are, on observable academic preparation, weaker than the failers who persist into the classroom despite failing. The gap in Panel A is therefore more likely to understate than overstate the difference between narrow passers and the teachers a lower bar would let in, insofar as measures of academic ability are positively, if weakly, correlated with teacher effectiveness (Clotfelter et al., 2007; Rockoff et al., 2011). The exam therefore screens in two senses. Its score predicts effectiveness better than the general pedagogy exam, and its threshold on average excludes candidates who would raise reading achievement less than those it admits.

5.4 Can Preparation Lower the Barrier?

The results so far establish that the exam is a meaningful barrier to teacher supply and also a real signal of teacher effectiveness. One response may be to lower the threshold. An alternative is to help more candidates clear it. In this section, I test whether Texas Reading Academies implemented at EPPs can do this.

I focus on the ten EPPs that adopted academies in 2024–25, the first year they were offered, comparing their first-attempt SOR outcomes to those of never-adopting programs before and after adoption. A second wave adopted in 2025–26, but I exclude them from the main analysis for two reasons. First, they fail parallel trends, entering on a declining performance path relative to non-adopters in the years before adoption (Table A12). Second, their post-adoption window covers only part of a school year (data are only available through March 2026). These programs lack a quality comparison group and have an insufficient amount of outcome data.

For the first cohort, by contrast, the pre-period trends are clean (Figure 5). Academy and non-academy programs moved together on both SOR pass rates and scores before adoption, and the pre-period event-study coefficients are individually and jointly insignificant (Table 11; joint $p = 0.529$ for pass rates, $p = 0.544$ for scores). The two groups then diverge in 2024–25. Academy programs post a 4.0 percentage point gain in first-attempt SOR pass rates (column 1) and a 0.15 standard deviation gain in scores (column 2), with difference-in-differences estimates of similar size (Panel B). The gap holds into the partial 2025–26 year, at 5.4 percentage points and 0.18 standard deviations, so the advantage does not fade as the first candidates move through.

To check whether the gain is specific to SOR content, I turn to the PPR exam, which includes content the academies do not cover (Figure 6). Literacy-specific preparation should raise SOR performance more than PPR, and the results are mixed. The PPR pass rate rises 4.2 percentage points in the adoption year (column 3), close to the SOR gain, but falls to a null in the partial 2025–26 year, and the pooled difference-in-differences estimates are smaller than the SOR gains and insignificant (Panel B, columns 3 and 4). The PPR movement is thus confined to the first year rather than sustained, which cuts against the

reading that adopting programs were simply improving across the board. Some spillover is plausible in any case, since coursework on explicit, systematic instruction may sharpen general pedagogical skill.

Taken together, the results suggest the academies help exam performance, but they fall short of proving it. Candidates at EPPs in the first-cohort of academy adopters improved on the SOR exam, but a similar first-year gain on the PPR exam means I cannot fully rule out that these programs were improving in more general ways. And the second cohort, which lacks clean pre-trends and a full year of data, cannot corroborate the first. What the design can establish is that adopting programs were on parallel trends beforehand and pulled ahead on the tested content after, which is consistent with the academies working.

6 Discussion

Occupational licensing may operate as a *screen* on quality but can also act as a *barrier* and restrict supply (M. M. Kleiner, 2000). Applied to teacher certification exams, prior work has largely found that they raise significant barriers and provide little value as screens. Angrist and Guryan (2008) showed that states which introduced stricter certification requirements reduced teacher supply and raised wages without altering teacher quality, and Deneault et al. (2025) found that making a Texas certification exam harder did not shift the distribution of teacher value-added. The natural prior for a new certification exam is therefore that its cost as a barrier will outweigh its benefit as a screen. The results here do not overturn that prior, but they show that it does not apply uniformly. Texas' Science of Reading exam is a substantial barrier to teacher supply. It also carries information about teacher effectiveness that the general pedagogy exam does not, and the candidates its threshold excludes are, on average, modestly less effective at raising reading achievement than the ones it admits.

The SOR exam acts as a barrier at two margins. Narrowly failing delays certification and reduces the probability of teaching the following year. This is noteworthy in a context where 15.6 percent of teachers enter the classroom with no certification at all (Ghazzawi et al., 2025), since it shows the exam binds even where the credential is not strictly required. The cost varies by certification route in a way prior literature has not documented. For

candidates close to certification, such as those finishing university-based programs or those already certified in another field or state, narrow failure is a delay. For alternatively certified candidates, the same failure cuts the probability of teaching the following October by twelve percentage points, because passing is what unlocks the internship through which they enter the classroom. This matters as an increasingly large share of new teachers come from alternative routes (National Council on Teacher Quality, 2025).

The market-level results extend the barrier to candidates further from the cutoff and to those who avoid the exam altogether. Prior work has documented the consequences of raising a passing standard or introducing testing where none existed, but not the effect of adding an exam to an existing certification pipeline. New certifications in fields that came to require the exam fell relative to comparison fields within the same programs and have not recovered five years later. The decline survives allowing pre-existing trends to continue into the post-period, though its statistical significance depends on the inference method, and two checks rule out substitution across fields. What these results do not show is what the decline meant for schools. Fewer new certifications in elementary and middle-grades fields does not by itself establish that districts could not fill positions, or that the teachers who filled them were worse than the candidates the exam deterred. Chronic shortages in Texas, as elsewhere, concentrate in mathematics, science, and special education (Cowan et al., 2016), none of which the requirement touches, so the affected fields had more slack to absorb the decline than a shortage field would. A state imposing the same requirement in a shortage field, or at a time when interest in teaching is falling faster than it already is (Bartanen & Kwok, 2023; Kraft & Lyon, 2024), would face a steeper tradeoff.

On the screening side, this study pushes back against the case for loosening or eliminating certification exams altogether (Hanushek & Rivkin, 2007). SOR scores predict student reading achievement two to four times more strongly than scores on the general pedagogy exam, and the pedagogy score adds nothing once the SOR score is known. The magnitude is modest, about 0.07 standard deviations of teacher effectiveness per standard deviation of score, but it is notable against two decades of research that have struggled to find pre-hire measures that predict teacher value-added at all (Clotfelter et al., 2007; Cowan et al., 2023; Goldhaber, 2007; Goldhaber et al., 2017a; T. J. Kane et al., 2008; Rockoff et al., 2011). In

this setting, an exam tied to a specific body of instructional knowledge is a better predictor of effectiveness than one testing general pedagogy. Whether that generalizes to other content exams, or reflects something particular to reading, is a question these data cannot answer.

The threshold also excludes candidates who, on average, would have taught reading less well than the ones it admits. First-attempt failers who reach the classroom anyway have reading value-added about a tenth of a standard deviation of teacher effectiveness below candidates who narrowly passed, and the failers who reach the classroom are positively selected on academic preparation relative to those who do not, so the gap for the full population of failers is likely larger. This is consistent with Larsen et al. (2026), who find that states tightening licensing requirements mostly removed candidates from the bottom of the quality distribution. If raising a requirement screens out weaker candidates, lowering one should admit them.

Two considerations complicate that arithmetic, in opposite directions. The first is that the analyses here observe only who is admitted, not what the requirement does to preparation. A pass/fail exam signals to candidates and programs what knowledge the state considers essential, and programs that reorganize coursework around it may produce better reading teachers among candidates who would have passed anyway. Candidates who learn the content to pass may teach reading better than they otherwise would have. And a school in which every early-grades teacher shares a foundation in reading instruction may benefit in ways that no individual teacher's value-added captures. These channels operate on candidates and programs I never observe being affected, and they could make the exam more valuable than the screening estimates suggest. The second consideration runs the other way. The cost of excluding a candidate depends on who replaces them. In Texas, a district of innovation may fill a vacancy with a teacher holding no certification at all (Anglin, 2024), and the evidence from pandemic-era emergency pathways is that such teachers are less effective than traditionally prepared peers (Backes et al., 2024). A requirement that turns away a candidate with substantial preparation but a score just below 240, and leaves the classroom to someone with no preparation, may lower reading instruction rather than protect it. The exam's net effect on the quality of instruction students receive therefore depends on labor market conditions that vary across districts and over time, and the estimates here speak to

the average candidate rather than to that margin.

Given the current climate, policymakers are likely to maintain minimum competency standards using certification exams. Roughly 40 percent of fourth graders score below basic on the National Assessment of Educational Progress, the highest share since 2002 (National Center for Education Statistics, 2025). Teacher quality is among the most consequential school-level determinants of student achievement (Chetty et al., 2014; Rivkin et al., 2005), certification requirements are among the few levers states directly control, and twenty states are now using them for exactly this purpose (National Council on Teacher Quality, 2024). The results here suggest that a well-designed exam can carry real information about teacher quality, but also that enforcing a standard through a pass/fail threshold alone is a costly way to use it.

The alternative this paper points to is doing more with the exam than using a cut score to determine eligibility for certification. The licensure literature has focused on the consequences of testing for entry and comparatively little on what programs and states can do to help candidates meet the standard, even though preparation is the one lever that preserves both supply and the requirement. The early evidence on Texas Reading Academies is consistent with structured preparation raising pass rates, though a similar first-year gain on the pedagogy exam and a second cohort that fails its own pre-trend test mean it should be read as suggestive rather than settled. Beyond preparation, the score itself is one of the stronger pre-hire signals of teaching quality available, and programs could use it to direct support toward candidates likely to struggle rather than discarding it once the pass/fail decision is made. Neither approach removes the minimum competency standard the exam creates, but both treat the exam as a source of information about teacher quality rather than only a gate to a credential.

6.1 Limitations

Several limitations bound these conclusions. The regression discontinuity estimates are local to the passing margin. The market-level analysis rests on a small number of clusters, its statistical significance depends on the inference method, and it cannot fully separate the SOR requirement from the concurrent redesign of the affected content exams or from

pandemic-era disruptions to the broader teacher labor market. The value-added estimates establish predictive validity rather than a causal effect of teacher knowledge on student achievement, since higher-scoring teachers may differ from lower-scoring peers in unobserved ways. The comparison of failers to narrow passers observes only failers who reached a classroom, and the characteristics on which I can assess their selection are weak predictors of teacher effectiveness, so the direction of bias is argued rather than established. The Reading Academies evidence comes from ten programs in a single adoption year. Finally, this is one exam in one state, in fields without chronic shortages, and the costs and screening value estimated here need not generalize to a different exam, a different subject, or a labor market with less slack.

7 Conclusion

Texas' Science of Reading exam affected the teacher pipeline in ways that matter for both supply and quality. The requirement imposed costs on candidates, especially those entering through alternative routes, and contributed to a persistent decline in new certifications in affected fields. But the exam also carries information that existing certification exams do not. Its score predicts reading value-added where the general pedagogy exam does not, and the candidates its threshold excludes are, on average, less effective at raising reading achievement than those it admits. The tradeoff a state faces is therefore real on both sides, and the evidence here does not resolve it in general, since the net effect depends on who fills the positions the exam leaves open and on channels, such as changes in how programs prepare candidates, that these data cannot observe. What the evidence does suggest is that the tradeoff is not fixed. Early results from Texas Reading Academies indicate that structured preparation can raise pass rates, which points to a way for states to hold a standard without accepting its full supply cost. As more states adopt Science of Reading requirements, the evidence from Texas argues for pairing them with investments in preparation that help candidates meet them.

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Tables and Figures

Table 1: Summary Statistics for First-Attempt Science of Reading Exam Takers

	Full Sample			Passed			Failed		
	Mean	SD	*N*	Mean	SD	*N*	Mean	SD	*N*
Passed SOR Exam	0.805	0.396	21692	1.000	0.000	17459	0.000	0.000	4233
Scaled Score on SOR Exam	251.6	17.3	21692	258.1	10.3	17459	224.8	14.3	4233
Certification Route									
Traditional	0.343	0.475	21692	0.354	0.478	17459	0.294	0.456	4233
Alternative	0.395	0.489	21692	0.363	0.481	17459	0.527	0.499	4233
By Exam	0.111	0.314	21692	0.116	0.320	17459	0.089	0.285	4233
Out of State	0.152	0.359	21692	0.167	0.373	17459	0.089	0.285	4233
Teaching before SOR Exam	0.315	0.465	21643	0.316	0.465	17419	0.311	0.463	4224
Demographics									
Has demographic information	0.645	0.479	21692	0.633	0.482	17459	0.692	0.462	4233
Female	0.921	0.269	13986	0.930	0.256	11056	0.890	0.313	2930
Non-white	0.429	0.495	14503	0.402	0.490	11496	0.530	0.499	3007
Outcomes									
Retake Exam	0.142	0.349	21692	0.000	0.008	17459	0.726	0.446	4233
Earn Standard Cert. in Six Months	0.432	0.495	21692	0.487	0.500	17459	0.204	0.403	4233
Earn Standard Cert. in One Year	0.588	0.492	21692	0.657	0.475	17459	0.306	0.461	4233
Employed as a Teacher next October	0.708	0.455	21692	0.738	0.439	17459	0.581	0.499	4233

Notes: Sample includes all first-attempt takers of the TExES Science of Reading exam (exam 293) from January 1, 2021 through October 23, 2023, restricted to candidates with valid scaled scores. Passed and Failed columns split the sample by first-attempt pass status (score $\geq$ 240). Demographic information is available for candidates who can be linked to PEIMS K–12 student records or THECB postsecondary enrollment records; the reduced sample sizes for Female and Non-white reflect this linkage. Certification outcomes measure whether the candidate earns any standard teaching certificate within the specified window. Employment is measured as of the last Friday in October following the exam date.

Table 2: Descriptive Statistics for New Certifications

	Pre-Mandate		Post-Mandate	
	Mean	N	Mean	N
SOR-Required Fields	0.417	139,575	0.360	203,241
Non-SOR Fields	0.583	139,575	0.640	203,241
Route				
Traditional	0.240	139,575	0.227	203,241
Alternative	0.338	139,575	0.335	203,241
By Exam	0.323	139,575	0.302	203,241
Out of State	0.099	139,575	0.136	203,241
Demographics				
Female	0.747	139,575	0.757	203,241
White	0.592	139,575	0.552	203,241
Hispanic	0.251	139,575	0.276	203,241
Black	0.108	139,575	0.117	203,241

Notes: Table reports descriptive statistics for the certification-level sample underlying the market-level analysis, prior to collapsing to the educator preparation program (EPP) × program-type × certification-field × half-year panel. Sample includes all newly issued standalone teaching certifications in the SOR-required and comparison fields listed in Table A2, 2016 through 2025. Post-mandate is defined as certifications issued from 2019H2 onward.

Table 3: Structure of the Market-Level Analytic Panel

	SOR-Required	Comparison
Certification fields	3	16
EPP-program-type units	200	200
Panel cells (EPP × field × half-year)	9,585	51,120
Share of cells zero-filled	51.2%	69.9%
Total certifications in panel	131,349	211,467

Notes: This table describes the panel formed by collapsing certification records in Table 2 to counts by educator preparation program (EPP) × program-type × certification-field × half-year, spanning 2016H1 through 2025H2. Certification fields are the 19 individual certification categories classified as SOR-required or comparison fields listed in Table A2. EPP-program-type units are the distinct combinations of educator preparation program and program type (e.g., Traditional, Alternative) observed issuing certifications. Panel cells are EPP-program-type × certification-field × half-year observations; a cell is coded zero if the EPP-program-type issued no certifications in that field during that half-year but was otherwise active, defined as having issued a certification in any field during that half-year. Periods before an EPP-program-type's first issued certification, or after its last, are excluded from the panel rather than coded as zero.

Table 4: Pre-Period EPP Characteristics by Texas Reading Academy Status

	Academy EPPs		Non-Academy EPPs		Difference	
	Mean	SD	Mean	SD	Mean	SE
Certification Exams						
N Exam Takers	298.6	266.0	469.9	1616.2	171.3	(513.9)
Pass SOR Exam	0.871	0.171	0.908	0.080	0.037	(0.030)
SOR Exam Scaled Score	252.9	10.4	252.8	7.0	-0.027	(2.442)
Pass PPR Exam	0.860	0.095	0.862	0.142	0.002	(0.046)
PPR Exam Scaled Score	259.0	8.3	258.8	8.3	-0.181	(2.757)
Program Type						
Traditional	0.733	0.411	0.558	0.472	-0.175	(0.155)
Alternative	0.267	0.411	0.412	0.467	0.145	(0.154)
By Exam	0.000	0.000	0.025	0.150	0.025	(0.048)
Out of State	0.000	0.000	0.005	0.051	0.005	(0.016)
Certs Issued						
Certs. Requiring SOR Exam	0.491	0.090	0.509	0.171	0.018	(0.055)
Certs. Not Requiring SOR Exam	0.509	0.090	0.491	0.171	-0.018	(0.055)
Demographics						
Female	0.810	0.075	0.784	0.098	-0.026	(0.032)
Asian	0.013	0.008	0.023	0.028	0.010	(0.009)
Black	0.047	0.040	0.121	0.193	0.073	(0.061)
Other	0.022	0.010	0.023	0.017	0.001	(0.005)
Hispanic	0.427	0.295	0.405	0.257	-0.023	(0.087)
White	0.490	0.267	0.429	0.248	-0.062	(0.083)
N EPPs	10		98		108	

Notes: Table compares pre-period characteristics of Educator Preparation Programs (EPPs) authorized to offer Texas Reading Academy coursework beginning in 2024–25 to EPPs that never adopted Texas Reading Academies during the study period. Means are calculated over academic years 2021–22 through 2023–24 (pre-academy period). The difference column reports the non-academy minus academy difference with standard errors from a two-sample t-test in parentheses. EPPs that adopted Texas Reading Academies in 2025–26 are excluded from both groups to avoid contamination from anticipatory changes at programs preparing to adopt. See Table A11 for the full list of EPPs and adoption years.

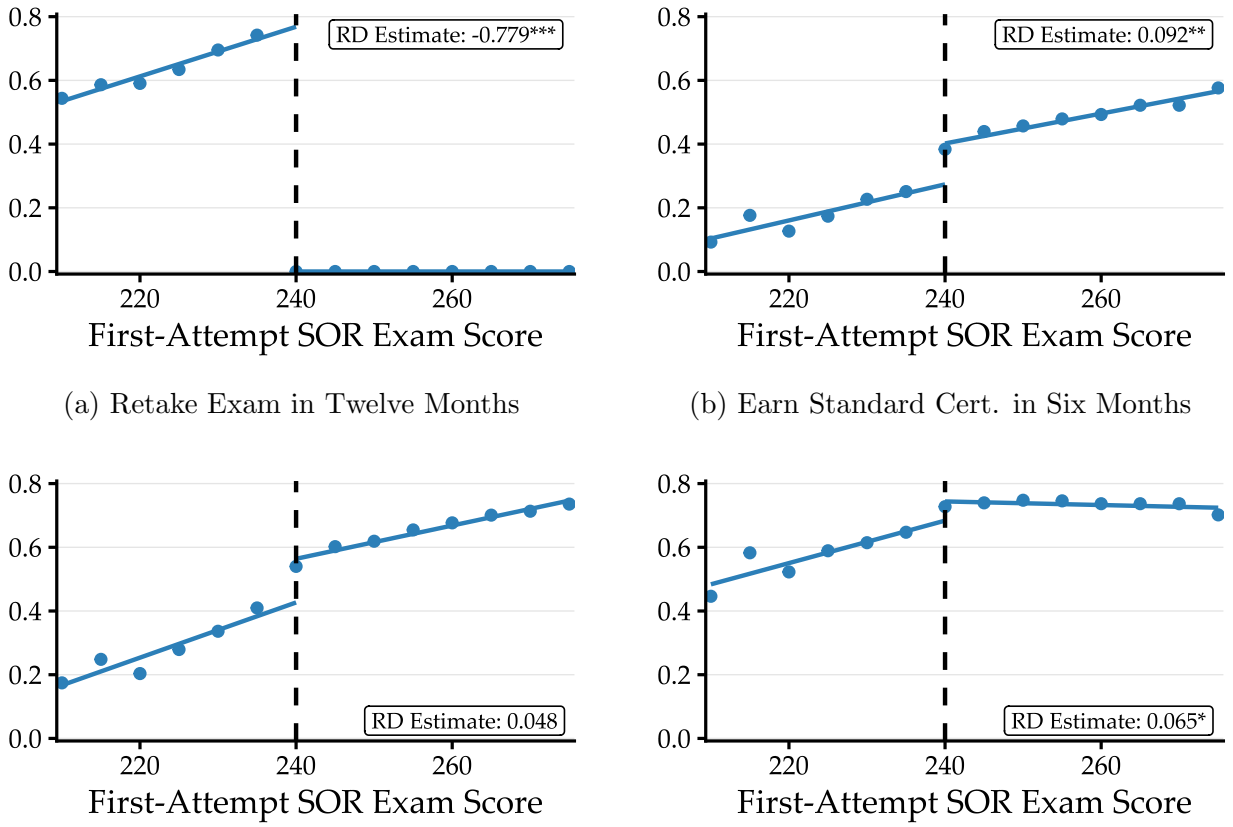
† Demographics presented for individuals who earn Standard teaching certificates associated with an EPP.

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table 5: RD Estimates of Passing the SOR Exam

	(1) Retake SOR Exam	(2) Earn Std. Cert. in 6 Months	(3) Earn Std. Cert. in 1 Year	(4) Teach Next October
Conventional	-0.777^{***}	0.0932^{***}	0.0592^{*}	0.0650^{*}
	(0.0191)	(0.0242)	(0.0266)	(0.025)
Bias-corrected	-0.779^{***}	0.0923^{***}	0.0485	0.0660^{**}
	(0.0191)	(0.0242)	(0.0266)	(0.025)
Robust	-0.779^{***}	0.0923^{**}	0.0485	0.0660^{*}
	(0.023)	(0.0293)	(0.0301)	(0.0304)
Month of Exam FE	✓	✓	✓	✓
N	21,692	21,692	21,692	21,692

Notes: Table reports regression discontinuity estimates of the effect of passing the Science of Teaching Reading (SOR) exam (TExES 293) on teacher outcomes. Each column presents a separate outcome. Columns (1) shows the probability of retaking the SOR exam within one year of the exam date. Columns (2) and (3) show the probability of earning a standard teaching certificate within six months and one year of the exam date, respectively. Column (4) shows the probability of working as a teacher in Texas public schools on the last Friday of October following the exam date. All specifications include month-of-exam fixed effects. All estimates use a local linear regression with triangular kernel weights and data-driven bandwidth selection. Standard errors in parentheses. Conventional and bias-corrected rows report heteroskedasticity-robust standard errors; robust rows report the bias-corrected robust standard errors of Calonico et al. (2014), which account for bias in the local polynomial estimator near the boundary. Sample includes all first-attempt test takers of the SOR exam from January 2021 to October 2023.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$



(a) Retake Exam in Twelve Months

(b) Earn Standard Cert. in Six Months

(c) Earn Standard Cert. in Twelve Months

(d) Work as a Teacher Next October

Figure 1: Teacher Outcomes by Science of Reading Exam Score

Notes: Each panel plots outcomes for first-attempt test takers of the TExES Science of Reading (SOR) exam (exam 293) from January 1, 2021 to October 23, 2023. X-axis shows first-attempt exam scores grouped into five-point bins (e.g., scores 220–224 are assigned to bin 220); points represent the mean outcome proportion within each bin. Lines show separate linear fits on each side of the passing threshold of 240 (indicated by the vertical dashed line). Panel (a) shows the proportion who retake the SOR exam within one year. Panel (b) shows the proportion who earn a Standard teaching certificate within six months of the exam date. Panel (c) shows the proportion who earn a Standard teaching certificate within twelve months of the exam date. Panel (d) shows the proportion who work as a teacher in Texas public schools on the last Friday of October following the exam date. To protect individual privacy, cells with proportions below one percent are coded as zero and cells above 95 percent are coded as one. The annotated RD estimate in each panel is the bias-corrected robust regression discontinuity estimate of Calonico et al. (2014), using a local linear regression with triangular kernel weights and data-driven bandwidth selection, corresponding to the robust estimates reported in Table 5.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

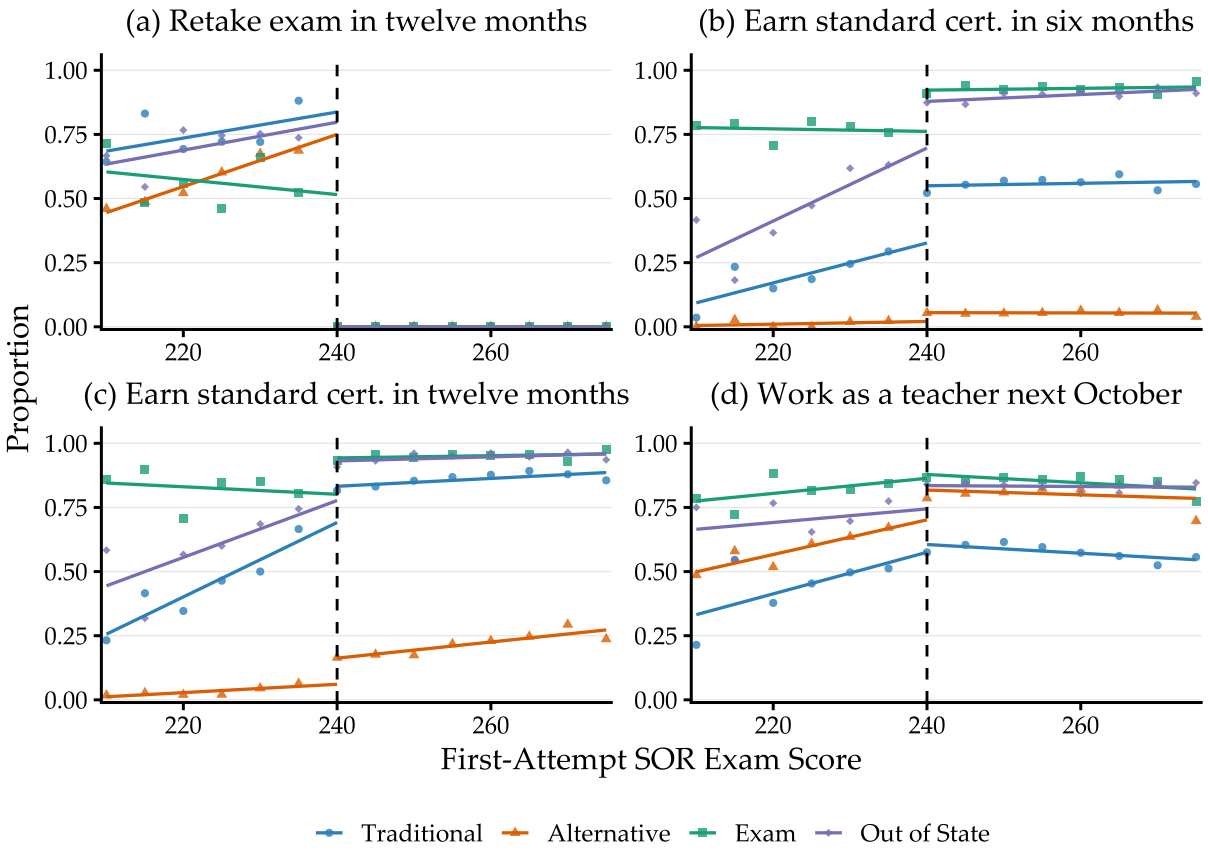


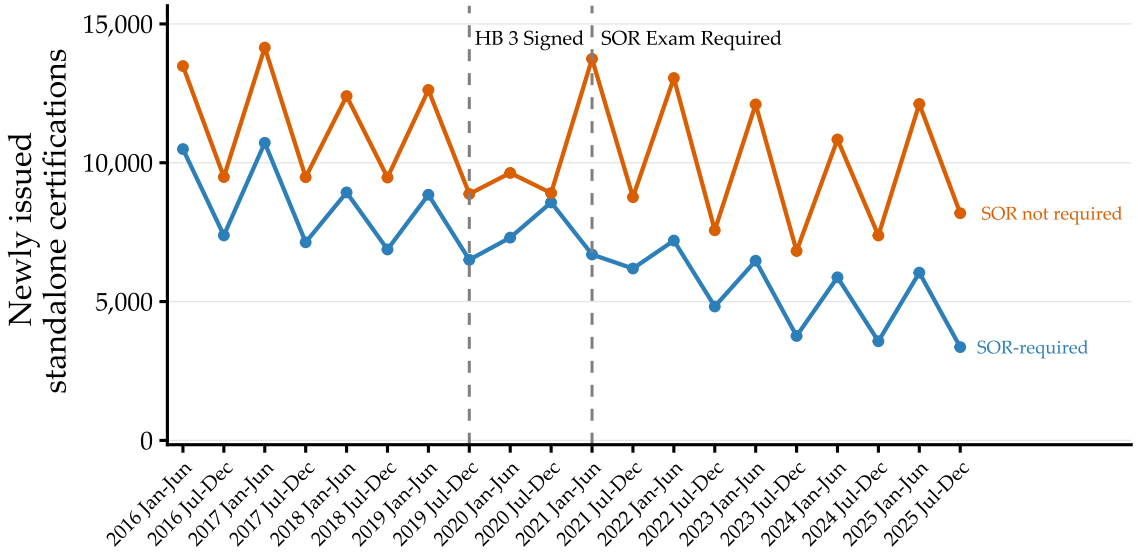
Figure 2: Teacher Outcomes by SOR Exam Score and Certification Route

Notes: Each panel plots outcomes by certification route for first-attempt test takers of the TExES Science of Reading (SOR) exam (exam 293) from January 1, 2021 to October 23, 2023. Certification routes are Traditional (university-based educator preparation programs), Alternative (alternative certification providers), By Exam (candidates certified by examination without a preparation program), and Out of State (candidates with out-of-state credentials). X-axis shows first-attempt exam scores grouped into five-point bins (e.g., scores 220–224 are assigned to bin 220); points represent the mean outcome proportion within each bin. Lines show separate linear fits on each side of the passing threshold of 240 (indicated by the vertical dashed line). Panel (a) shows the proportion who retake the SOR exam within one year. Panel (b) shows the proportion who earn a Standard teaching certificate within six months of the exam date. Panel (c) shows the proportion who earn a Standard teaching certificate within twelve months of the exam date. Panel (d) shows the proportion who work as a teacher in Texas public schools on the last Friday of October following the exam date. To protect individual privacy, cells with proportions below one percent are coded as zero and cells above 95 percent are coded as one.

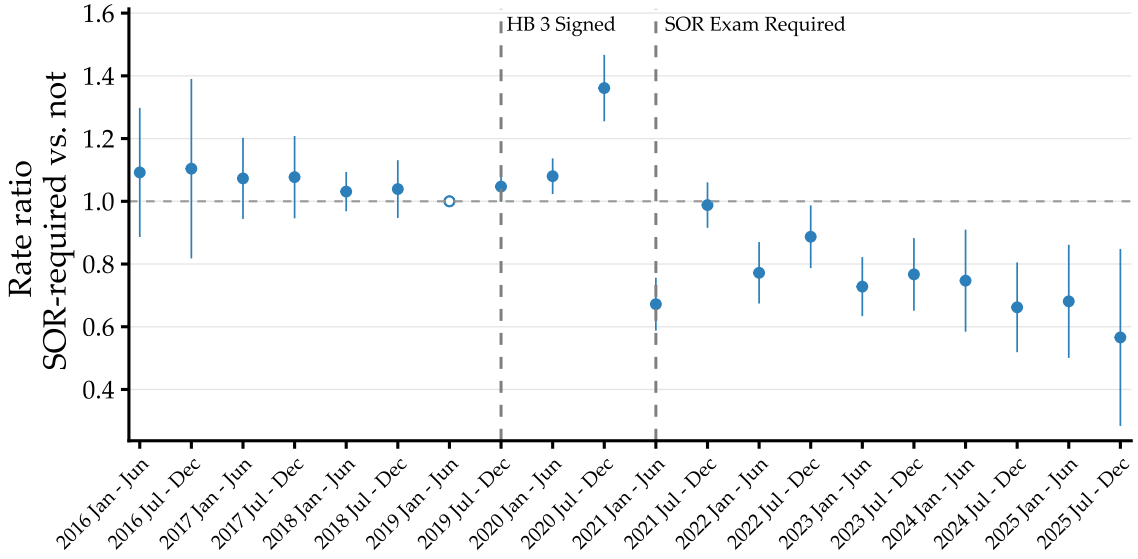
Table 6: RD Estimates of Passing the SOR Exam, by Certification Route

	(1) Retake SOR Exam	(2) Earn Std. Cert. in 6 Months	(3) Earn Std. Cert. in 1 Year	(4) Teach Next October
Traditional	-0.924^{***}	0.160^{***}	-0.017	0.0267
	(0.0398)	(0.048)	(0.058)	(0.0533)
Alternative	-0.681^{***}	0.025	0.106^{***}	0.120^{**}
	(0.0388)	(0.014)	(0.025)	(0.0443)
By Exam	-0.553^{***}	0.160	0.139	0.0281
	(0.0844)	(0.103)	(0.095)	(0.0908)
Out of State	-0.781^{***}	0.218^{**}	0.097	-0.0442
	(0.0598)	(0.073)	(0.068)	(0.0719)

Notes: Table reports regression discontinuity estimates of the effect of passing the Science of Teaching Reading (SOR) exam (TExES 293) on teacher outcomes, estimated separately for each certification route. Each cell reports the estimate from a separate regression. Column (1) shows the probability of retaking the SOR exam within one year. Columns (2) and (3) show the probability of earning a standard teaching certificate within six months and one year of the exam date, respectively. Column (4) shows the probability of working as a teacher in Texas public schools on the last Friday of October following the exam date. All specifications include month-of-exam fixed effects, which compare candidates who sat for the exam at a similar point relative to the October employment snapshot; they are retained for the certification outcomes for comparability and have little effect on the estimates (Table A3). Estimates use local linear regression with triangular kernel weights and data-driven bandwidth selection. Estimates are bias-corrected and standard errors are the robust standard errors of Calonico et al. (2014), reported in parentheses. Traditional route includes candidates enrolled in university-based educator preparation programs; Alternative includes alternative certification providers; By Exam includes candidates who satisfied certification requirements by examination; Out of State includes candidates with credentials from another state. Sample includes first-attempt SOR exam takers from January 2021 to October 2023.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$



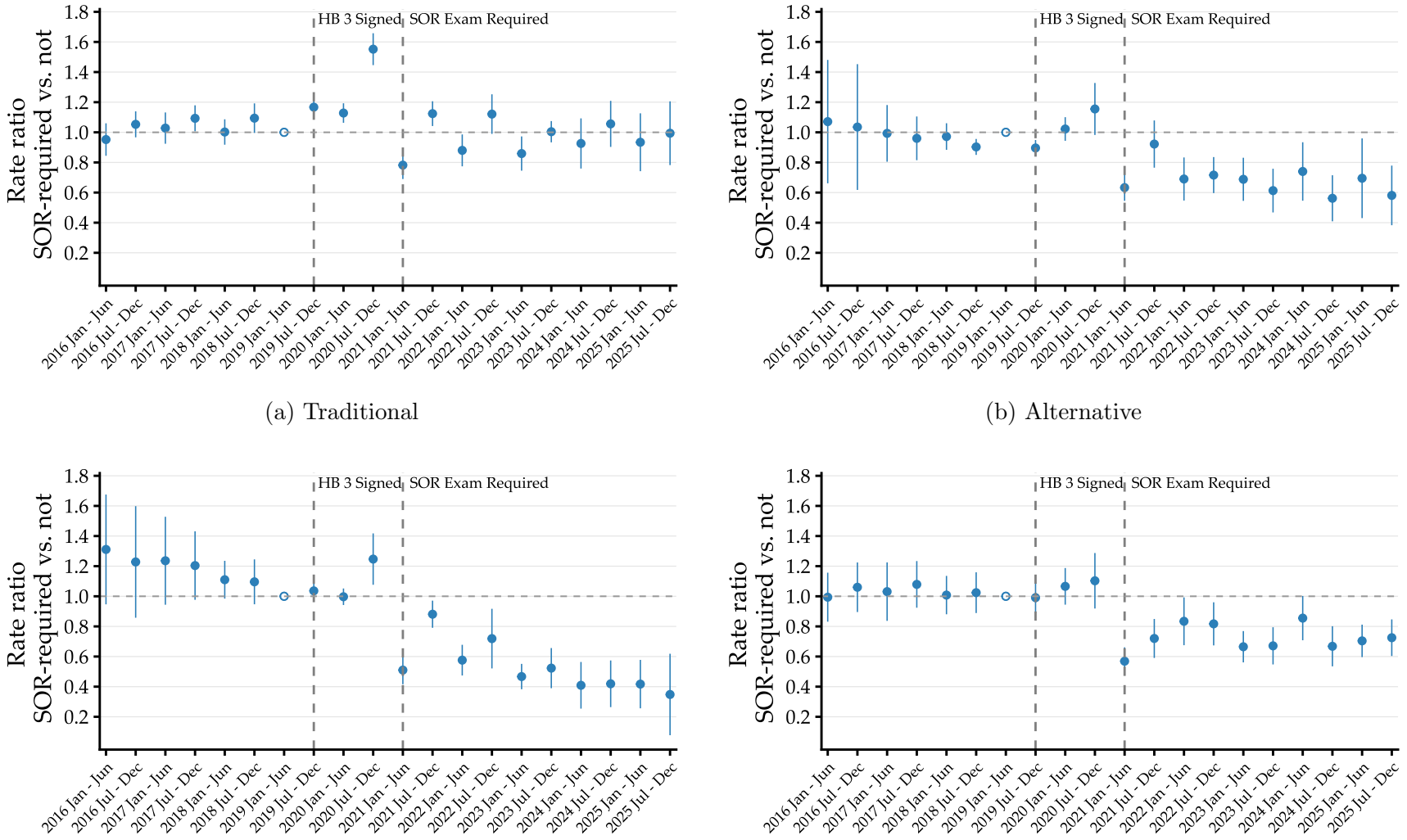
(a) Semi-annual certification counts



(b) Event study estimates

Figure 3: New Certifications in SOR-Required Fields

Notes: Panel (a) plots the number of newly issued standalone Texas teaching certifications requiring versus not requiring the TExES Science of Teaching Reading (SOR) exam (TExES 293), by half-year. The dotted line indicates when HB 3 was signed into law (June 2019); the dashed line indicates when the SOR requirement took effect (January 2021). Panel (b) plots coefficients from a difference-in-differences event study estimated by Poisson pseudo-maximum likelihood on data collapsed to the educator preparation program (EPP) × program type × half-year × certification type level, where the outcome is the count of newly issued certifications. Each point gives the rate ratio of certifications in SOR-required relative to non-SOR fields, normalized to the omitted period, 2019 January–June (hollow marker). Specifications include EPP × program type, certification type, and half-year fixed effects. Vertical bars give 95% confidence intervals; standard errors are clustered at the certification type level. Table A2 lists SOR-required and comparison fields. A newly issued certification refers to the first time an individual receives that certification; non-teaching and supplemental credentials are excluded.



(a) Traditional

(b) Alternative

(c) By Exam

(d) Out of State

Figure 4: New Certifications in SOR-Required Fields Relative to 2019H1, by Certification Route

Notes: Each panel reproduces the market-level difference-in-differences event study of Figure 3 for a single certification route: Traditional (panel a), Alternative (panel b), Certification by Exam (panel c), and Out of State (panel d). Within each route, coefficients are estimated by Poisson pseudo-maximum likelihood on data collapsed to the educator preparation program (EPP) × program type × half-year × certification type level, where the outcome is the count of newly issued certifications. Each point gives the rate ratio of certifications in SOR-required relative to non-SOR fields in that half-year, normalized to the omitted period, 2019 January–June (hollow marker). The dotted vertical line indicates when HB 3 was signed into law (June 2019); the dashed vertical line indicates when the SOR requirement took effect (January 2021). Specifications include EPP × program type, certification type, and half-year fixed effects. Vertical bars give 95% confidence intervals; standard errors are clustered at the certification type level. Table A2 lists SOR-required and comparison fields. A newly issued certification refers to the first time an individual receives that certification; non-teaching and supplemental credentials are excluded.

Table 7: Effect of the SOR Requirement on New Certifications: Event Study

	All	Traditional	Alternative	By Exam	Out of State
	(1)	(2)	(3)	(4)	(5)
2016 Jan–Jun × SOR Req.	1.092	0.952	1.071	1.311	0.994
	(0.105)	(0.055)	(0.209)	(0.186)	(0.083)
2016 Jul–Dec × SOR Req.	1.104	1.053	1.035	1.228	1.060
	(0.146)	(0.044)	(0.213)	(0.189)	(0.084)
2017 Jan–Jun × SOR Req.	1.073	1.028	0.993	1.236	1.031
	(0.066)	(0.053)	(0.096)	(0.149)	(0.099)
2017 Jul–Dec × SOR Req.	1.077	1.093*	0.960	1.204	1.079
	(0.067)	(0.044)	(0.074)	(0.116)	(0.079)
2018 Jan–Jun × SOR Req.	1.031	1.002	0.972	1.110	1.008
	(0.032)	(0.043)	(0.045)	(0.064)	(0.065)
2018 Jul–Dec × SOR Req.	1.039	1.094*	0.903***	1.096	1.024
	(0.047)	(0.050)	(0.027)	(0.076)	(0.069)
2019 Jan–Jun × SOR Req.	1.000	1.000	1.000	1.000	1.000
2019 Jul–Dec × SOR Req.	1.047	1.167***	0.896***	1.036	0.991
	(0.025)	(0.041)	(0.027)	(0.037)	(0.050)
2020 Jan–Jun × SOR Req.	1.080**	1.128***	1.022	0.997	1.066
	(0.029)	(0.033)	(0.040)	(0.028)	(0.062)
2020 Jul–Dec × SOR Req.	1.361***	1.552***	1.155	1.247**	1.103
	(0.054)	(0.054)	(0.088)	(0.087)	(0.094)
2021 Jan–Jun × SOR Req.	0.672***	0.782***	0.633***	0.510***	0.569***
	(0.043)	(0.047)	(0.045)	(0.048)	(0.047)
2021 Jul–Dec × SOR Req.	0.988	1.124**	0.922	0.881*	0.720***
	(0.037)	(0.042)	(0.080)	(0.046)	(0.066)
2022 Jan–Jun × SOR Req.	0.772***	0.880*	0.690***	0.576***	0.834
	(0.050)	(0.054)	(0.073)	(0.052)	(0.081)
2022 Jul–Dec × SOR Req.	0.887*	1.121	0.716***	0.719*	0.817*
	(0.051)	(0.067)	(0.061)	(0.101)	(0.073)
2023 Jan–Jun × SOR Req.	0.728***	0.859*	0.688***	0.467***	0.665***
	(0.048)	(0.058)	(0.073)	(0.043)	(0.053)
2023 Jul–Dec × SOR Req.	0.767***	1.004	0.613***	0.523***	0.671***
	(0.059)	(0.036)	(0.074)	(0.068)	(0.063)
2024 Jan–Jun × SOR Req.	0.747**	0.926	0.740*	0.409***	0.855
	(0.083)	(0.085)	(0.099)	(0.079)	(0.075)
2024 Jul–Dec × SOR Req.	0.662***	1.056	0.562***	0.419***	0.668***
	(0.073)	(0.078)	(0.078)	(0.079)	(0.068)
2025 Jan–Jun × SOR Req.	0.681**	0.934	0.695	0.417***	0.704***
	(0.092)	(0.098)	(0.135)	(0.082)	(0.055)
2025 Jul–Dec × SOR Req.	0.566*	0.994	0.581**	0.348**	0.725***
	(0.144)	(0.108)	(0.101)	(0.138)	(0.062)
N	60,705	28,367	31,426	380	532
Joint F-test of pre-period (p-value)	0.638	< 0.001	< 0.001	0.009	0.290

Notes: Table reports estimates from Poisson pseudo-maximum likelihood regressions of newly issued standalone teaching certifications on an interaction between certification-field SOR-required status and time. All specifications include EPP × program-type, certification-field, and half-year fixed effects. Coefficients are normalized to the omitted reference period, 2019 January–June. Each point-estimate row gives the rate ratio of certifications in SOR-required relative to non-SOR fields in that half-year. Column (1) uses the full sample; columns (2)–(5) restrict to the traditional, alternative, certification-by-examination, and out-of-state routes, respectively. Coefficients are exponentiated and reported as rate ratios. Standard errors clustered at the certification-field level are provided in parenthesis.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table 8: Effect of the SOR Requirement on New Certifications: Pooled Estimates

	All (1)	Traditional (2)	Alternative (3)	By Exam (4)	Out of State (5)
Trend-adjusted level shift[a]	0.822	0.972	0.827	0.604	0.764
Clustered SE p-value	[0.002]	[0.630]	[0.004]	[< 0.001]	[< 0.001]
RI p-value[b]	[0.096]	[0.750]	[0.114]	[0.043]	[0.059]
Post × SOR Req. (2019H2–2025H2)	0.785	0.994	0.780	0.530	0.760
Clustered SE p-value	[0.016]	[0.941]	[0.133]	[< 0.001]	[0.005]
RI p-value[b]	[0.128]	[0.955]	[0.367]	[0.033]	[0.250]
Donut: Post × SOR Req. (2022H1–2025H2)[c]	0.686	0.931	0.677	0.403	0.726
Clustered SE p-value	[0.004]	[0.463]	[0.054]	[< 0.001]	[0.002]
RI p-value[b]	[0.107]	[0.695]	[0.239]	[0.022]	[0.237]
N	60,705	28,367	31,426	380	532

Notes: Table reports pooled estimates from Poisson pseudo-maximum likelihood regressions of newly issued standalone teaching certifications on the interaction between certification-field SOR-required status and time. All specifications include EPP × program-type, certification-field, and half-year fixed effects. Coefficients are exponentiated and reported as rate ratios. Column (1) uses the full sample; columns (2)–(5) restrict to the traditional, alternative, certification-by-examination, and out-of-state routes. Each estimate is followed by the p-value from the certification-field clustered standard error and the randomization inference p-value, both reported in brackets.

[a]The trend-adjusted specification includes a differential linear trend for SOR-required fields estimated on pre-announcement half-years only (2016H1–2019H1). The counterfactual for the post period is the level that fitted trend reaches in 2019H1, so the reported level shift measures the average post-announcement gap relative to the endpoint of the pre-mandate trend rather than to the average pre-announcement differential.

[b]Randomization inference p-value is calculated by re-estimating the specification 2,000 times, each time randomly assigning SOR-required status to 3 of the 19 certification fields while holding the true number of treated fields fixed, and computing the t-statistic on the coefficient of interest in each placebo estimate. The reported p-value is the share of these placebo t-statistics whose magnitude equals or exceeds that of the t-statistic from the true assignment.

[c]The donut specification excludes half-years 2019H2 through 2021H2, removing the anticipatory certification surge preceding implementation and its immediate aftermath.

Table 9: Predicting Student Achievement with Teacher Certification Exam Scores

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A. Reading									
SOR Exam	0.011***	0.013***	0.013***				0.011**	0.009*	0.012***
	(0.002)	(0.002)	(0.002)				(0.004)	(0.003)	(0.004)
PPR Exam				0.003	0.005***	0.005***	−0.003	−0.001	−0.006
				(0.002)	(0.001)	(0.001)	(0.004)	(0.003)	(0.004)
N	829,964	829,964	829,964	1,440,994	1,440,994	1,440,994	395,669	395,669	395,669
N Teachers	8,301	8,301	8,301	14,576	14,576	14,576	4,987	4,987	4,987
Panel B. Math									
SOR Exam	0.015***	0.017***	0.014***				0.020***	0.017***	0.016**
	(0.003)	(0.003)	(0.003)				(0.006)	(0.005)	(0.005)
PPR Exam				0.006*	0.007***	0.010***	−0.003	−0.002	0.007
				(0.002)	(0.002)	(0.002)	(0.006)	(0.005)	(0.005)
N	808,593	808,593	808,593	1,367,248	1,367,248	1,367,248	385,255	385,255	385,255
N Teachers	8,292	8,292	8,292	14,447	14,447	14,447	4,979	4,979	4,979
Grade FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
District FE		✓			✓			✓	
Campus FE			✓			✓			✓

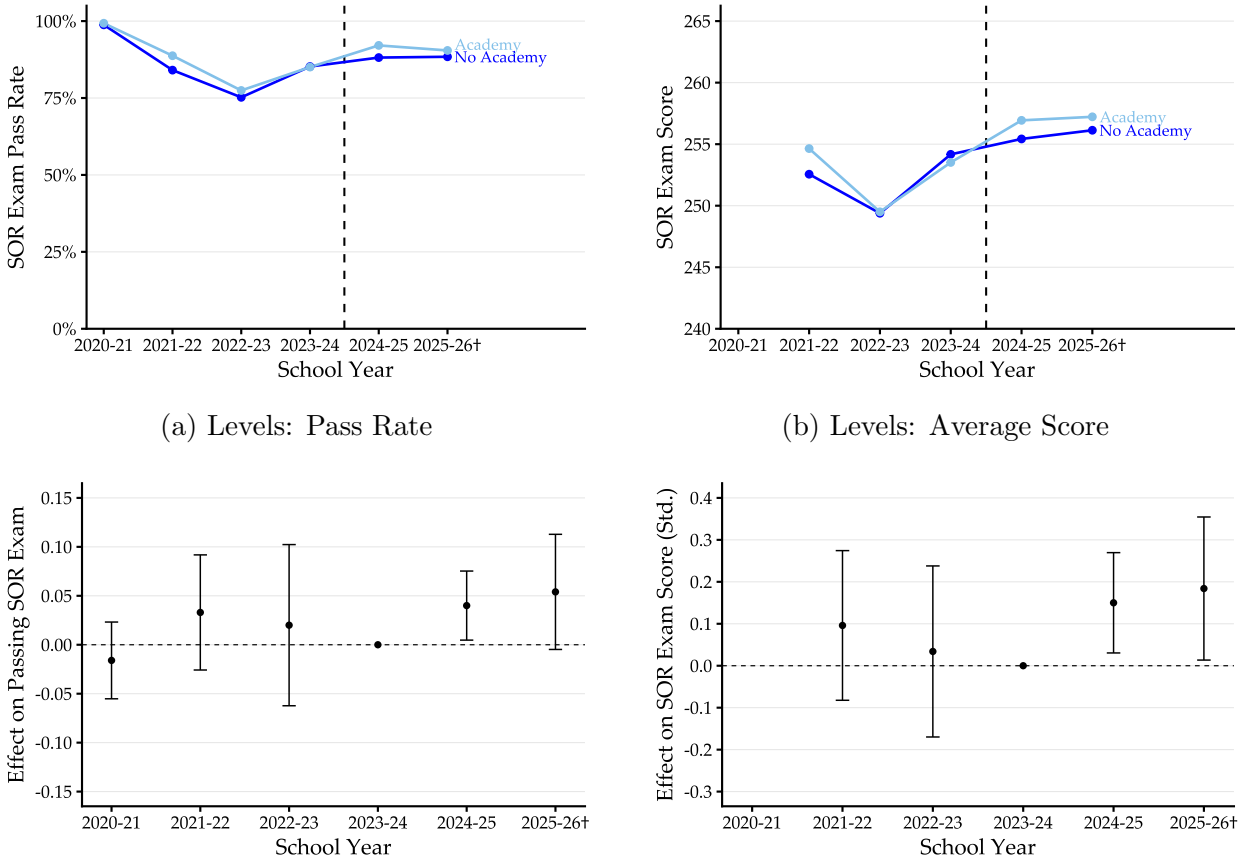
Notes: Table reports estimates from value-added models estimated via ordinary least squares (OLS) relating teachers' standardized certification exam scores to student achievement on the State of Texas Assessments of Academic Readiness (STAAR). Panel A reports student reading achievement, and Panel B reports student math achievement. The dependent variable is the corresponding student test score standardized to have mean zero and standard deviation one within grade, year, and subject. Each column reports a separate regression. Teacher certification exam scores are standardized to have mean zero and standard deviation one among test takers. Columns (1)–(3) estimate the association between Science of Reading (SOR) exam scores and student achievement, columns (4)–(6) use Pedagogy and Professional Responsibilities (PPR) exam scores, and columns (7)–(9) enter both scores jointly, so that each coefficient reflects the association net of the other. All specifications include controls for prior reading and math achievement (quadratic and cubic terms), race, gender, special education status, gifted status, economic disadvantage, and classroom- and school-level averages of these covariates, along with grade and year fixed effects. Columns (2), (5), and (8) add district fixed effects, and columns (3), (6), and (9) add campus fixed effects; the fixed-effects rows are reported once, in Panel B, and apply to both panels. Standard errors clustered at the teacher level are reported in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table 10: Value-Added Below the Passing Threshold and Selection into Having Value-Added Estimates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Failing the exam, teacher effectiveness, and who is measured								
	Reading value-added				Observed with reading value-added			
First-attempt score range	230–250	220–250	210–250	200–250	230–250	220–250	210–250	200–250
Fail on first attempt	-0.017	-0.017	-0.016*	-0.021*	-0.015	-0.028**	-0.034***	-0.038***
	(0.010)	(0.009)	(0.008)	(0.008)	(0.011)	(0.010)	(0.009)	(0.009)
% of takers below floor	4.91	2.40	1.31	0.64	4.91	2.40	1.31	0.64
N	1,818	2,051	2,142	2,193	6,909	7,996	8,469	8,761
Panel B. Who appears in the value-added sample, among first-attempt failers								
	Female	White	Fam. inc. < $60k	SAT score	Female	White	Fam. inc. < $60k	SAT score
Observed with reading VA	-0.033*	0.003	-0.064*	24.980*	-0.034*	0.003	-0.062	20.781*
	(0.015)	(0.022)	(0.032)	(10.621)	(0.015)	(0.022)	(0.032)	(10.406)
First-attempt score					0.001	-0.000	-0.001	1.812***
					(0.000)	(0.001)	(0.001)	(0.391)
N	2,930	3,007	1,434	872	2,930	3,007	1,434	872

Notes: Each column reports a separate regression estimated via OLS. In Panel A the sample is candidates whose first-attempt Science of Teaching Reading (TExES 293) score falls in the range given in the column header. Columns (1)–(4) regress estimated reading value-added on an indicator for failing on the first attempt (i.e., a first-attempt score below 240). Value-added is the teacher fixed effect from a student-level model with the controls listed in the notes to Table 9 and no certification exam scores. Columns (5)–(8) use the same four windows and replace the outcome with an indicator for being observed with reading value-added. The percent of takers below floor is the percentage of all first-attempt takers scoring below the bottom of each range. In Panel B the sample is limited to all first-attempt failers. Each column regresses the listed characteristic on an indicator for being observed with reading value-added, with certification-route fixed effects. Columns (5)–(8) repeat the specification adding the first-attempt score as a control. Family income and SAT score are observed only for candidates who can be linked to Texas Higher Education Coordinating Board records, which accounts for the varying samples in those columns. Heteroskedasticity-robust standard errors reported in parentheses throughout.

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$



(a) Levels: Pass Rate

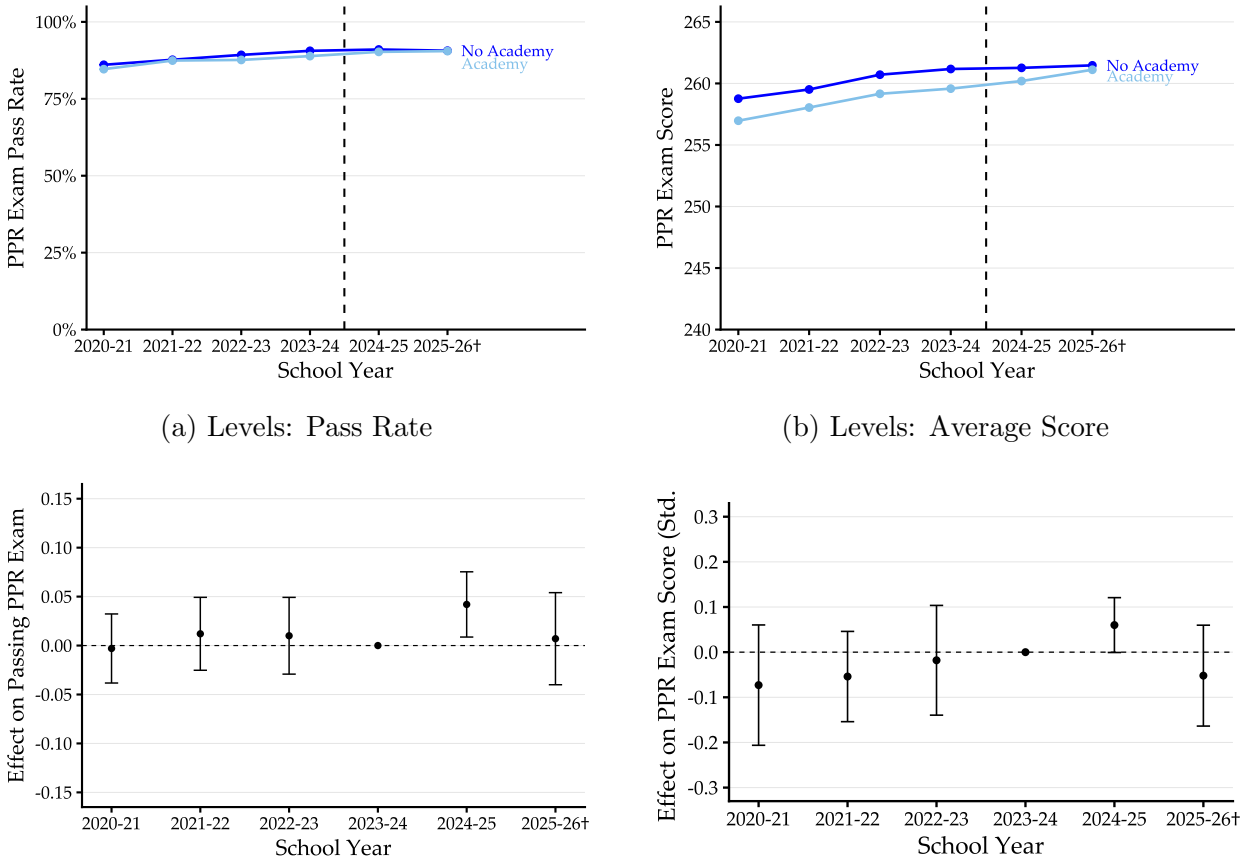
(b) Levels: Average Score

(c) Event Study: Pass Rate

(d) Event Study: Average Score

Figure 5: TExES Science of Reading (SOR) Outcomes by Texas Reading Academy Adoption

Notes: Panels (a) and (b) show levels of pass rates and average scores, respectively, for the Science of Reading (SOR) exam, comparing EPPs with and without Texas Reading Academies. Panels (c) and (d) show event study estimates for the same outcomes, with the reference year (2023–24) indicated by a hollow marker. Texas Reading Academies were first implemented in the 2024–25 school year. Pass rate is measured as the proportion of first-attempt test takers who pass the exam (score $\geq$ 240). Average score represents the mean exam score among all first-attempt test takers, standardized to have mean zero and standard deviation of one within each year. † Certification exam data through March 2026.



(a) Levels: Pass Rate

(b) Levels: Average Score

(c) Event Study: Pass Rate

(d) Event Study: Average Score

Figure 6: Pedagogy and Professional Responsibilities (PPR) Outcomes by Texas Reading Academy Adoption

Notes: Panels (a) and (b) show levels of pass rates and average scores, respectively, for the Pedagogy and Professional Responsibilities (PPR) exam, comparing EPPs with and without Texas Reading Academies. Panels (c) and (d) show event study estimates for the same outcomes, with the reference year (2023–24) indicated by a hollow marker. Texas Reading Academies were first implemented in the 2024–25 school year. Pass rate is measured as the proportion of first-attempt test takers who pass the PPR exam. Average score represents the mean exam score among all first-attempt test takers, standardized to have mean zero and standard deviation of one within each year. 2020–21 scaled scores are unavailable due to piloting. † Certification exam data through March 2026.

Table 11: Effect of Texas Reading Academies on Certification Exam Outcomes

	SOR		PPR	
	Pass Rate (1)	Score (2)	Pass Rate (3)	Score (4)
Panel A. Event Study				
2020–21 × Texas Reading Academy	−0.016		−0.003	−0.073
	(0.020)		(0.018)	(0.068)
2021–22 × Texas Reading Academy	0.033	0.096	0.012	−0.054
	(0.030)	(0.091)	(0.019)	(0.051)
2022–23 × Texas Reading Academy	0.020	0.034	0.010	−0.018
	(0.042)	(0.104)	(0.020)	(0.062)
2023–24 × Texas Reading Academy	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)
2024–25 × Texas Reading Academy	0.040*	0.150*	0.042*	0.060+
	(0.018)	(0.061)	(0.017)	(0.031)
2025–26 × Texas Reading Academy†	0.054+	0.184*	0.007	−0.052
	(0.030)	(0.087)	(0.024)	(0.057)
Panel B. Difference-in-Differences				
Post × Texas Reading Academy	0.038+	0.119	0.027	0.061
	(0.023)	(0.096)	(0.019)	(0.046)
Joint F-test of pre-period	0.529	0.544	0.638	0.642
N	61,446	40,756	95,617	95,617
Pre-period mean, non-Academy	0.856	0.029	0.854	-0.004
Pre-period mean, Academy	0.861	0.007	0.851	-0.015

Notes: Table reports estimates from event study (Panel A) and difference-in-differences (Panel B) specifications comparing certification exam outcomes at Educator Preparation Programs (EPPs) that adopted Texas Reading Academies in 2024–25 to EPPs that never adopted. Regressions are estimated on first-attempt exam records, so EPPs are weighted by their number of test takers. The outcomes are an indicator for passing on the first attempt and the standardized first-attempt score on the SOR exam (the TExES Science of Teaching Reading exam, exam 293) and the Pedagogy and Professional Responsibilities (PPR) exam. The PPR sample includes all first-attempt PPR takers at each EPP, including candidates in certification fields that do not require the SOR exam. All specifications include EPP fixed effects and year fixed effects. The reference year for the event study is 2023–24, the year immediately preceding academy implementation. The joint F-test reports the p-value from a test that all pre-period interaction coefficients are jointly zero. EPPs that adopted Texas Reading Academies in 2025–26 are excluded from both groups. † Certification exam data through March 1, 2026. Standard errors clustered at the Educator Preparation Program level in parentheses. 2020–21 scaled scores are unavailable due to piloting.

$^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Appendix A. Additional Tables and Figures

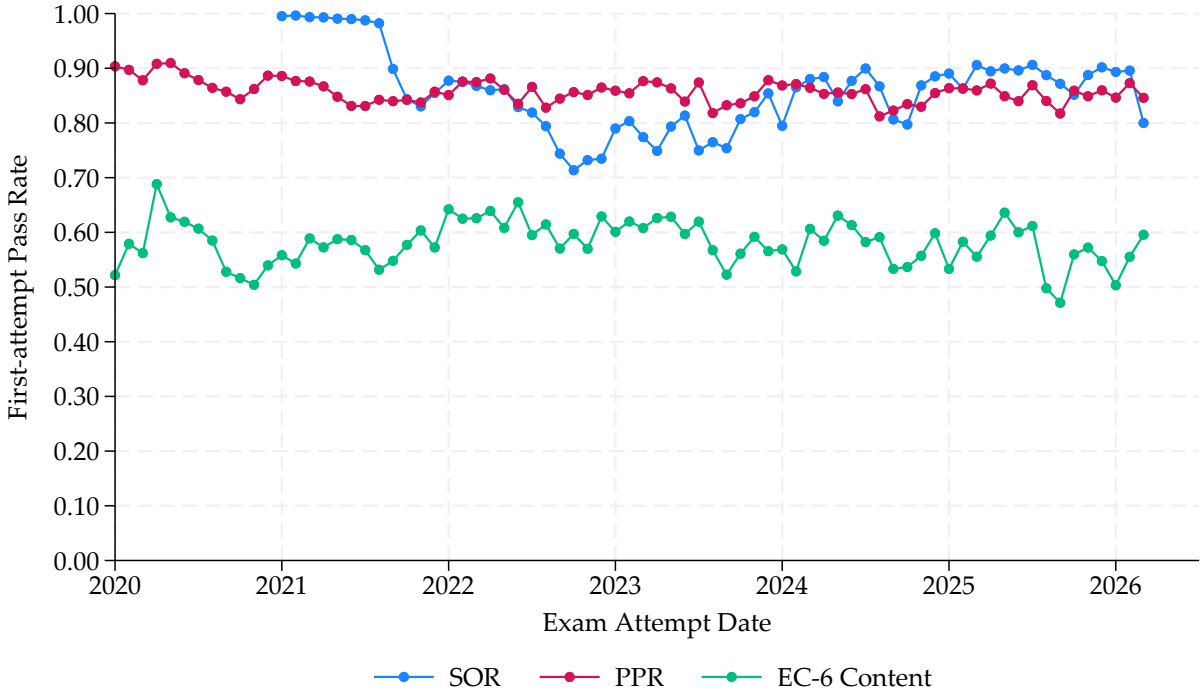


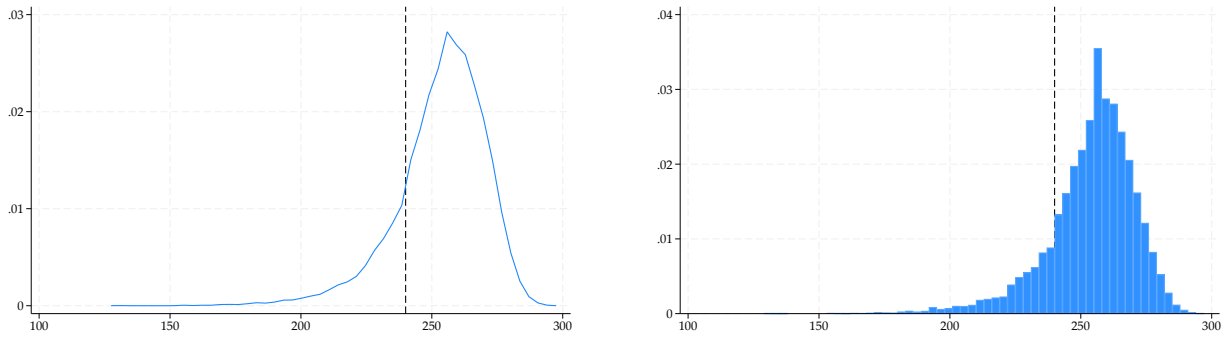
Figure A1: First-Attempt Pass Rates for Texas Teacher EC-6 Certification Exams

Notes: This figure plots first-attempt pass rates for three Texas teacher certification exams from 2020 to 2025. The blue line shows pass rates for the Science of Reading (SOR) exam (exam 293). The orange line shows pass rates for the Pedagogy and Professional Responsibilities (PPR) exam. The green line shows pass rates for the EC-6 Content exam. The sample includes all first-attempt test takers in Texas. The black vertical line indicates January 1, 2021, when the SOR exam was introduced. The red dashed vertical line indicates when the State Board for Educator Certification (SBEC) began including scaled scores from the SOR exam after initial piloting.

Table A1: Covariates at the SOR Exam Passing Threshold

	Conventional	Bias-corrected	Robust
Female	0.00574	0.0122	0.0122
	(0.0202)	(0.0202)	(0.0241)
White	0.0326	0.0385	0.0385
	(0.0321)	(0.0321)	(0.0385)
Traditional	0.000748	−0.00555	−0.00555
	(0.0218)	(0.0218)	(0.0253)
Alternative	0.0112	0.00795	0.00795
	(0.0269)	(0.0269)	(0.0323)
By Exam	0.0106	0.0168	0.0168
	(0.0166)	(0.0166)	(0.0194)
Out of State	−0.0168	−0.0209	−0.0209
	(0.0160)	(0.0160)	(0.0189)
Teaches Before Exam	−0.00896	−0.00950	−0.00950
	(0.0262)	(0.0262)	(0.0320)

Notes: Table reports regression discontinuity estimates at the SOR exam passing threshold (score ≥ 240) with each pre-determined covariate as the outcome. None of the estimates are statistically significant at the 0.05 level, supporting the assumption that candidates just above and below the cutoff are comparable in observable characteristics. All estimates use a local linear regression with triangular kernel weights and data-driven bandwidth selection. Heteroskedasticity-robust standard errors are reported in parentheses. Demographic variables (i.e., Female, White) are available only for candidates linked to PEIMS or THECB records.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$



(a) Density Plot

(b) Histogram

Figure A2: Distribution of Science of Reading Exam Scores

Notes: This figure plots the distribution of scores on the Texas Science of Reading (SOR) exam (exam 293) for first-attempt test takers between January 1, 2021 and October 23, 2023. Panel (a) shows a kernel density estimate of the score distribution. Panel (b) shows a histogram with overlaid density estimate. The vertical dotted line indicates the passing threshold of 240. A density discontinuity test using the method of Cattaneo et al. (2018) fails to reject the null hypothesis of no manipulation around the cutoff ($p = 0.051$), suggesting no evidence of sorting around the passing threshold.

Table A2: Classification of Texas Teaching Certificates by SOR-Exam Requirement

Certification Group	SOR (293) Required?
Panel A: SOR-required certifications	
Core Subjects: Early Childhood–Grade 6	Yes
Core Subjects: Grades 4–8	Yes
English Language Arts and Reading: Grades 4–8	Yes
Panel B: Non-SOR certifications (comparison group)	
Computer Science: Grades 8–12	No
English Language Arts and Reading: Grades 7–12	No
Fine Arts: EC–12	No
Foreign Language: EC–12	No
Health & Physical Education	No
Mathematics: Grades 4–8	No
Mathematics: Grades 7–12	No
Science: Grades 4–8	No
Science: Grades 6–12	No
Science: Grades 7–12	No
Social Studies: Grades 4–8	No
Social Studies: Grades 7–12	No
Special Education: EC–12	No
Vocational Education 6–12	No
Vocational Education 8–12	No
Vocational Education EC–12	No

Notes: Table lists the 19 certification groups in the market-level analysis sample, which are also the units at which standard errors are clustered. Classification follows the Texas Education Agency Required Test Chart (May 2024); a certification is coded SOR-required if the TExES Science of Teaching Reading exam (293) appears among its required content-pedagogy tests. Successive versions of a certificate are grouped into a single field, so that each SOR-required group is observed before and after the mandate: Core Subjects and Core Subjects with STR (and the earlier Generalist certificates) are grouped by grade band, Early Childhood: PK–3 is grouped with Core Subjects: EC–6, and English Language Arts and Reading/Social Studies: Grades 4–8 (with or without STR) is grouped with English Language Arts and Reading: Grades 4–8. Non-teaching credentials (Reading Specialist, Counselor, Diagnostician, Librarian, Principal, Superintendent) and supplemental certificates that attach to a base certificate rather than certifying a teaching field independently (Bilingual, ESL, Gifted & Talented, Special Education Supplemental) are excluded from the sample.

Table A3: RD Estimates of Passing the Science of Reading Exam: With and Without Controls

	Retake SOR Exam		Earn Std. Cert. in 6 Months		Earn Std. Cert. in 1 Year		Teach Next October	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Conventional	-0.777^{***}	-0.776^{***}	0.0932^{***}	0.0922^{***}	0.0592^{*}	0.0596^{*}	0.0650^{*}	0.0590^{*}
	(0.0191)	(0.0193)	(0.0242)	(0.0246)	(0.0266)	(0.0267)	(0.025)	(0.0258)
Bias-corrected	-0.779^{***}	-0.778^{***}	0.0923^{***}	0.0913^{***}	0.0485	0.0494	0.0660^{**}	0.0610^{*}
	(0.0191)	(0.0193)	(0.0242)	(0.0246)	(0.0266)	(0.0267)	(0.025)	(0.0258)
Robust	-0.779^{***}	-0.778^{***}	0.0923^{**}	0.0913^{**}	0.0485	0.0494	0.0660^{*}	0.061
	(0.023)	(0.0233)	(0.0293)	(0.0297)	(0.0301)	(0.0303)	(0.0304)	(0.0314)
Month of Exam FE	✓		✓		✓		✓	
N	21,692	21,692	21,692	21,692	21,692	21,692	21,692	21,692

Notes: Table reports regression discontinuity estimates of the effect of passing the Science of Teaching Reading (SOR) exam (TExES 293) on teacher outcomes with and without month-of-exam fixed effects. Odd-numbered columns include month-of-exam fixed effects; even-numbered columns do not. Columns (1)–(2) show the probability of retaking the SOR exam within one year of the exam date. Columns (3)–(4) show the probability of earning a Standard teaching certificate within six months of the exam date. Columns (5)–(6) show the probability of earning a Standard teaching certificate within one year of the exam date. Columns (7)–(8) show the probability of working as a teacher in Texas public schools on the last Friday of October following the exam date. Month-of-exam fixed effects account for the mechanical relationship between exam timing and the annual October employment snapshot. All estimates use a local linear regression with triangular kernel weights and data-driven bandwidth selection. Standard errors in parentheses. Conventional and bias-corrected rows report heteroskedasticity-robust standard errors; robust rows report the bias-corrected robust standard errors of Calonico et al. (2014), which account for bias in the local polynomial estimator near the boundary. Sample includes all first-attempt test takers of the SOR exam from January 2021 to October 2023.

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table A4: Robustness of RD Estimates to Bandwidth Selection Method

	Retake SOR Exam			Earn Std. Cert. 6 Mo.			Earn Std. Cert. 1 Yr.			Teach Next October		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Conventional	-0.777^{***}	-0.777^{***}	-0.772^{***}	0.0932^{***}	0.0922^{***}	0.0769^{*}	0.0592^{*}	0.0593^{*}	0.0307	0.0640^{*}	0.0639^{*}	0.0640^{*}
	(0.0191)	(0.0192)	(0.0252)	(0.0242)	(0.0253)	(0.0316)	(0.0266)	(0.0266)	(0.0347)	(0.0250)	(0.0250)	(0.0327)
Bias-corrected	-0.779^{***}	-0.779^{***}	-0.773^{***}	0.0923^{***}	0.0924^{***}	0.0772^{*}	0.0485	0.0492	0.0263	0.0650^{**}	0.0652^{**}	0.0647^{*}
	(0.0191)	(0.0192)	(0.0252)	(0.0242)	(0.0253)	(0.0316)	(0.0266)	(0.0266)	(0.0347)	(0.0250)	(0.0250)	(0.0327)
Robust	-0.779^{***}	-0.779^{***}	-0.773^{***}	0.0923^{**}	0.0924^{**}	0.0772^{*}	0.0485	0.0492	0.0263	0.0650^{*}	0.0652^{*}	0.0647
	(0.0230)	(0.0231)	(0.0269)	(0.0293)	(0.0302)	(0.0338)	(0.0301)	(0.0320)	(0.0362)	(0.0304)	(0.0298)	(0.0351)
BW Method	mserd	msesum	cerrd	mserd	cerrd	certwo	mserd	msesum	cerrd	mserd	msesum	cerrd
Bandwidth	13.4	13.3	8.1	13.2	12.0	8.0	13.0	13.1	7.9	12.3	12.3	7.5
Month of Exam FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
N	21,692	21,692	21,692	21,692	21,692	21,692	21,692	21,692	21,692	21,692	21,692	21,692

Notes: Table reports regression discontinuity estimates of the effect of passing the Science of Teaching Reading (SOR) exam under different data-driven bandwidth selection methods from Calonico et al. (2014). Each set of three columns presents a separate outcome estimated using three bandwidth methods: mserd (MSE-optimal, common bandwidth), msesum (MSE-optimal, sum of bandwidths), cerrd (CER-optimal, common bandwidth), and certwo (CER-optimal, two bandwidths). MSE-optimal bandwidths minimize mean squared error and tend to be wider; CER-optimal bandwidths ensure coverage error rate optimality and tend to be narrower. Employment specifications include month-of-exam fixed effects. Standard errors in parentheses. Conventional and bias-corrected rows report heteroskedasticity-robust standard errors; robust rows report the bias-corrected robust standard errors of Calonico et al. (2014), which account for bias in the local polynomial estimator near the boundary. Sample includes all first-attempt test takers of the SOR exam from January 2021 to October 2023.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table A5: Estimating Market-Level Results with OLS

	(1) Event study	(2) Full post	(3) Donut	(4) Pre-trend Adjusted
Panel A. Event study				
2016 Jan–Jun × SOR Req.	4.840			
	(2.915)			
2016 Jul–Dec × SOR Req.	−0.187			
	(2.267)			
2017 Jan–Jun × SOR Req.	5.325			
	(3.672)			
2017 Jul–Dec × SOR Req.	−0.838			
	(1.391)			
2018 Jan–Jun × SOR Req.	1.091			
	(0.785)			
2018 Jul–Dec × SOR Req.	−2.134			
	(2.099)			
2019 Jan–Jun × SOR Req.	0.000			
2019 Jul–Dec × SOR Req.	−2.872			
	(2.281)			
2020 Jan–Jun × SOR Req.	−1.804			
	(1.373)			
2020 Jul–Dec × SOR Req.	1.007*			
	(0.421)			
2021 Jan–Jun × SOR Req.	−4.422*			
	(1.382)			
2021 Jul–Dec × SOR Req.	−3.277*			
	(2.216)			
2022 Jan–Jun × SOR Req.	−2.726**			
	(0.288)			
2022 Jul–Dec × SOR Req.	−5.338**			
	(3.370)			
2023 Jan–Jun × SOR Req.	−4.042*			
	(1.294)			
2023 Jul–Dec × SOR Req.	−7.059**			
	(4.497)			
2024 Jan–Jun × SOR Req.	−4.307*			
	(1.253)			
2024 Jul–Dec × SOR Req.	−7.589**			
	(4.422)			
2025 Jan–Jun × SOR Req.	−4.225*			
	(0.800)			
2025 Jul–Dec × SOR Req.	−7.865**			
	(4.181)			
Panel B. Pooled specifications				
Post × SOR Req. (2019H2–2025H2)		−5.145**		
		(2.286)		
Donut: Post × SOR Req. (2022H1–2025H2)			−6.408**	
			(2.710)	
Trend-adjusted level shift				−3.340
				(1.223)
N	60,705	60,705	44,517	60,705
Joint *F*-test of pre-period (*p*-value)	0.090			

Notes: Table reproduces the market-level analysis by ordinary least squares as a check on the Poisson pseudo-maximum likelihood estimates reported in Table 7 and Table 8. The outcome is the count of newly issued standalone teaching certifications in levels, so coefficients are in certificates per EPP × program-type × certification-field × half-year. All specifications include EPP × program-type, certification-field, and half-year fixed effects. Column (1) reports the event study, normalized to the omitted reference period, 2019 January–June. Column (2) reports the pooled difference-in-differences estimate over the full post-announcement period. Column (3) reports the donut specification, which excludes half-years 2019H2 through 2021H2. Column (4) adds a linear trend in the SOR-required differential through the pre-announcement period and reports the post-announcement level shift relative to that trend. Standard errors in parentheses are wild cluster bootstrap standard errors with Webb weights and 9,999 replications, clustered at the certification-field level. The joint *F*-test reports the *p*-value from a test that the pre-announcement interaction coefficients are jointly zero.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table A6: Sensitivity of the Market-Level Estimate to Pre-Trend Violations

	2019 Jul–Dec to 2025 Jul–Dec		Donut-weighted: 2022 Jan–Jun to 2025 Jul–Dec	
	90% CI	95% CI	90% CI	95% CI
Original (unrestricted)	[0.730, 0.921]	[0.714, 0.941]	[0.614, 0.848]	[0.595, 0.874]
$M = 0.000$	[0.790, 0.997]	[0.772, 1.020]	[0.690, 0.937]	[0.670, 0.965]
$M = 0.001$	[0.773, 1.031]	[0.753, 1.057]	[0.668, 0.985]	[0.645, 1.017]
$M = 0.002$	[0.744, 1.093]	[0.722, 1.118]	[0.630, 1.069]	[0.606, 1.102]

Notes: Table reports HonestDiD confidence intervals, exponentiated and expressed as rate ratios, under the smoothness restriction Δ^{SD} (Rambachan & Roth, 2023) for the equally-weighted average post-mandate effect over two target windows: the full post-announcement period (2019H2–2025H2, including the anticipatory certification surge preceding the mandate's January 2021 implementation) and the post-implementation period (2022H1–2025H2). The donut-weighted column places zero weight on half-years 2019H2–2021H2 in the target average, rather than excluding them from the underlying event-study estimation. This allows the counterfactual pre-trend to run smoothly through the anticipatory and initial implementation periods before being evaluated against the post-implementation coefficients, and is more conservative than dropping these periods outright, since it lets any accumulated pre-trend deviation compound over more periods before reaching the target window. M is the maximum permitted period-to-period change in the counterfactual trend's slope, relative to a linear extrapolation of the pre-period trend; $M = 0$ imposes exact linear continuation.

Table A7: SOR Exam Performance and Characteristics of Students at First School

	Econ. Disadv.		LEP		SPED		Gifted		ESL	
	OLS	RD	OLS	RD	OLS	RD	OLS	RD	OLS	RD
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Estimate	-0.046^{***}	-0.009	-0.018^{***}	0.001	0.002^{***}	0.001	0.002^{***}	0.006	-0.002^{*}	-0.002
	(0.002)	(0.019)	(0.002)	(0.013)	(0.001)	(0.003)	(0.001)	(0.004)	(0.001)	(0.006)

Notes: Table reports the relationship between Science of Teaching Reading (SOR) exam performance and the characteristics of the students a candidate teaches in the first year following the exam. Each cell reports the estimate from a separate regression. Outcomes are the share of students at the candidate's school who are economically disadvantaged (1)–(2), limited English proficient (3)–(4), receiving special education services (5)–(6), identified as gifted and talented (7)–(8), and classified as English as a Second Language (9)–(10). Odd-numbered columns report OLS estimates of the association between a one standard deviation increase in the SOR exam score and each outcome. Even-numbered columns report regression discontinuity estimates of the effect of passing the SOR exam at the certification threshold, using local linear regression with triangular kernel weights and data-driven bandwidth selection; estimates are bias-corrected and standard errors are the robust standard errors of Calonico et al. (2014). All specifications include month-of-exam fixed effects. Standard errors reported in parentheses. Because student characteristics are observed only for candidates who go on to teach, these estimates are necessarily conditional on employment as a teacher in a Texas public school in the year following the exam, and the sample is correspondingly smaller than in the certification tables. Sample includes first-attempt SOR exam takers from January 2021 to October 2023.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table A8: RD Estimates of Passing the SOR Exam on Certification, by Certification Field

	(1)	(2)	(3)	(4)
	Earn Std. Cert. in 6 Months		Earn Std. Cert. in 1 Year	
	SOR Field	Non-SOR Field	SOR Field	Non-SOR Field
Conventional	0.112***	0.008	0.095***	−0.005
	(0.025)	(0.017)	(0.027)	(0.019)
Bias-corrected	0.106***	0.01	0.085**	−0.006
	(0.025)	(0.017)	(0.027)	(0.019)
Robust	0.106***	0.01	0.085**	−0.006
	(0.030)	(0.021)	(0.032)	(0.023)

Notes: Table reports regression discontinuity estimates of the effect of passing the Science of Teaching Reading (SOR) exam (TExES 293) on the probability of earning a Standard teaching certificate, estimated separately by certification field. Each cell reports the estimate from a separate regression. Columns (1) and (3) use as the outcome earning a certificate in a field for which the SOR exam is required; columns (2) and (4) use earning a certificate in a field for which it is not. SOR-required fields comprise early childhood and elementary generalist certifications (Core Subjects: Early Childhood–Grade 6), middle-grades generalist certifications (Core Subjects: Grades 4–8), and middle-grades English language arts and social studies certifications (English Language Arts and Reading: Grades 4–8; and English Language Arts, Reading/Social Studies: Grades 4–8). Because passing the SOR exam is a prerequisite only in the former group, the non-SOR columns serve as a falsification test. Columns (1) and (2) show the probability of earning a certificate within six months of the exam date; columns (3) and (4) within one year. All specifications include month-of-exam fixed effects. Estimates use local linear regression with triangular kernel weights and data-driven bandwidth selection. Conventional, bias-corrected, and robust estimates follow Calonico et al. (2014); standard errors reported in parentheses. Sample includes first-attempt SOR exam takers from January 2021 to October 2023.

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table A9: Appearing in the Value-Added Sample, Overall and Among Candidates Who Teach

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	All first-attempt takers				Teaching the following October			
First-attempt score range	230–250	220–250	210–250	200–250	230–250	220–250	210–250	200–250
Fail on first attempt	-0.015	-0.028**	-0.034***	-0.038***	0.011	0.006	0.002	0.000
	(0.011)	(0.010)	(0.009)	(0.009)	(0.015)	(0.013)	(0.013)	(0.012)
N	6,909	7,996	8,469	8,761	4,877	5,488	5,737	5,884

Notes: Columns (1)–(4) reproduce columns (5)–(8) of Panel A of Table 10. Columns (5)–(8) restrict the sample to candidates observed working as a teacher in a Texas public school in the October following their first attempt. The estimates in the restricted sample are small and statistically indistinguishable from zero throughout, which indicates that the gradient in the full sample reflects failing candidates not reaching a classroom at all rather than failing candidates who teach being sorted away from tested grades and subjects. Heteroskedasticity-robust standard errors reported in parentheses.

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table A10: Who Appears in the Value-Added Sample, Failers Versus the Comparison Group

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	First-attempt failers				Comparison group			
	Female	White	Fam. inc. < $60k	SAT score	Female	White	Fam. inc. < $60k	SAT score
Observed with reading VA	-0.033*	0.003	-0.064*	24.980*	-0.028**	0.031*	-0.037	28.337***
	(0.015)	(0.022)	(0.032)	(10.621)	(0.009)	(0.015)	(0.021)	(6.534)
N	2,930	3,007	1,434	872	6,246	6,431	3,175	2,076

Notes: Columns (1)–(4) reproduce columns (1)–(4) of Panel B of Table 10. Columns (5)–(8) estimate the same specification on the comparison group rather than on first-attempt failers. Each column regresses the listed characteristic on an indicator for being observed with reading value-added, with certification-route fixed effects, and the reported mean is the constant from that regression. What matters for bias in Panel A is not selection within either group but the difference between the two, since only differential selection across the two sides of the comparison shifts the estimated gap. Family income and SAT score are observed only for candidates who can be linked to Texas Higher Education Coordinating Board records. Heteroskedasticity-robust standard errors reported in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table A11: EPPs Offering Texas Reading Academies by Year of Adoption

Educator Preparation Program
2024–25 Academic Year
ESC19 Teacher Preparation & Certification Program
Lamar University
Teach Us Texas
Texas A&M University Commerce
Texas Christian University
Texas Tech University
Trinity University
University of Texas at El Paso (UTEP)
University of Texas at Tyler
West Texas A&M University
2025–26 Academic Year
240 Certification
ACTRGV/Pharr San Juan Alamo
Baylor University†
Dallas College
East Texas Baptist University
Houston Community College
LeTourneau University†
McMurry University
Region 4 Inspire Texas
Responsive Ed 180 Educator Preparation Program
Stephen F. Austin State University Department of Education Studies
Texas Woman's University
The University of Texas at Dallas - Teacher Development Center

Notes: Table lists Educator Preparation Programs (EPPs) authorized to offer Texas Reading Academies (TRA) coursework to preservice elementary teachers by year of adoption. TRA is a statewide professional development program that engages elementary educators with high-quality instructional strategies rooted in the Science of Reading (SOR). EPPs that adopted TRA in 2024–25 and continued in 2025–26 are listed only under 2024–25. Texas A&M University-Commerce was renamed East Texas A&M University in 2024, and the ESC19 Teacher Preparation & Certification Program appears on later TEA rosters as "Region 19"; both are 2024–25 adopters and are listed only under that year. Sources: 2024–25 cohort; 2025–26 cohort; current roster.

† Listed on TEA's 2025–26 roster as of September 2025 but since removed, indicating the program dropped TRA.

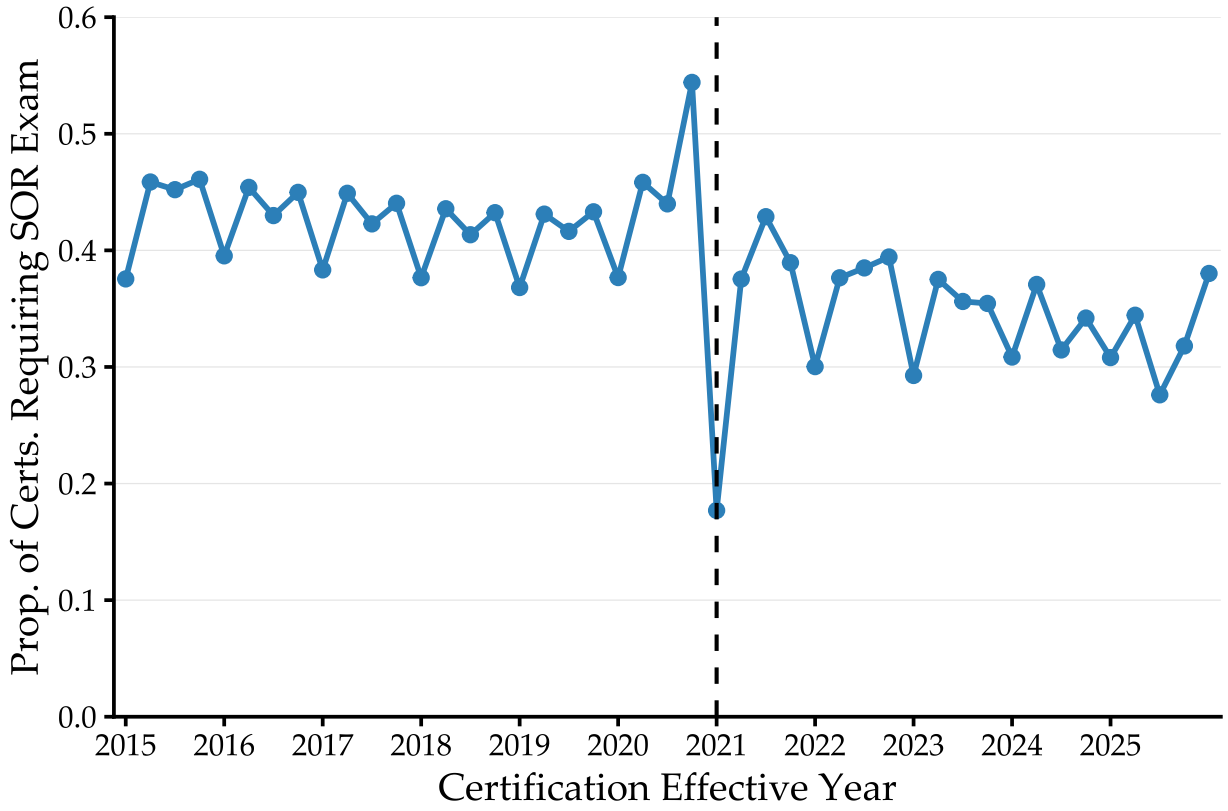
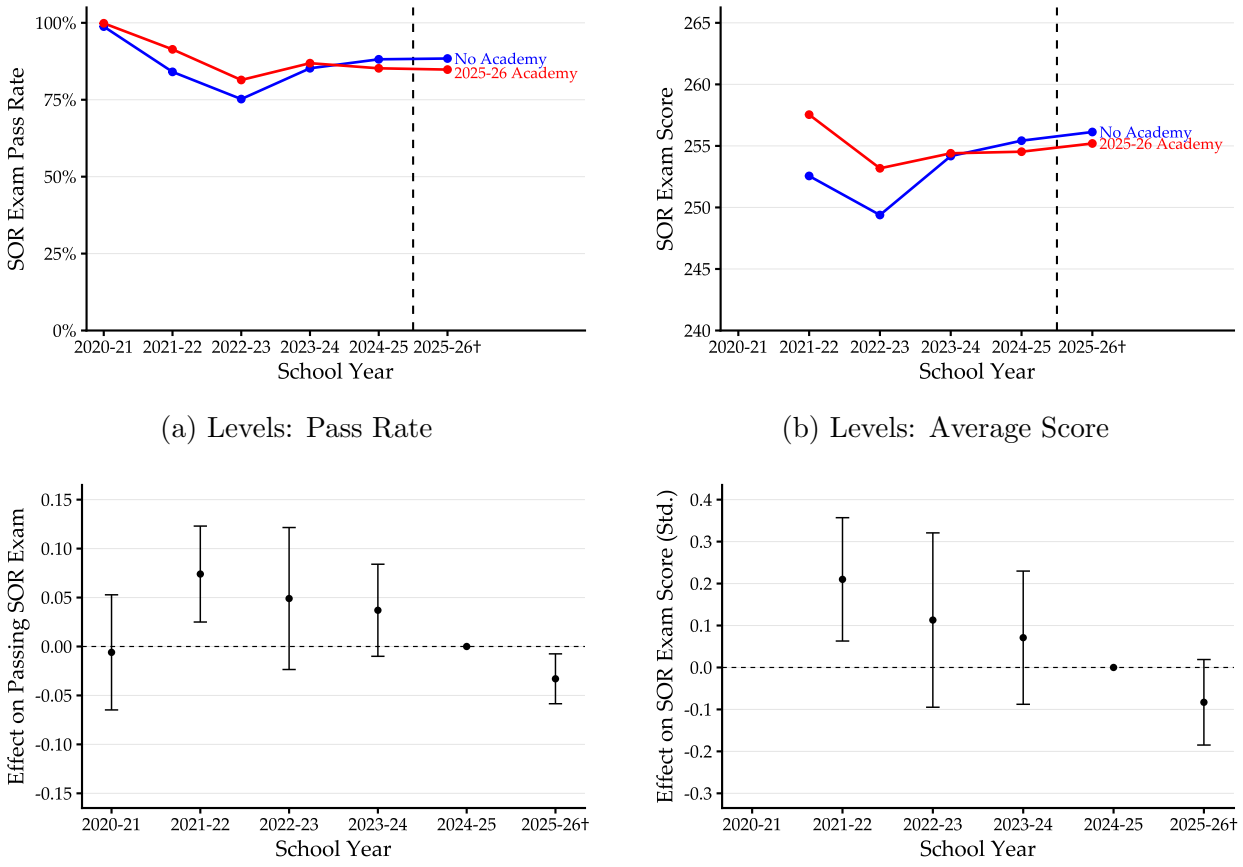


Figure A3: Proportion of New Teacher Certifications Requiring the Science of Reading Exam

Notes: The figure displays the proportion of newly issued Texas teacher certifications that require the TExES Science of Reading exam (exam 293) by quarter from 2018 through early 2025. Each point represents the quarterly average proportion across all certification types. The vertical dashed line indicates January 1, 2021, when the SOR exam requirement was implemented for certain certification types. A newly issued certification refers to the first time an individual receives that particular certification. The sample includes all teacher certification types issued in Texas during this period.



(a) Levels: Pass Rate

(b) Levels: Average Score

(c) Event Study: Pass Rate

(d) Event Study: Average Score

Figure A4: TExES Science of Reading (SOR) Outcomes by Texas Reading Academy Adoption, 2025–26 Cohort

Notes: Panels (a) and (b) show levels of pass rates and average scores, respectively, for the Science of Reading (SOR) exam, comparing EPPs with and without Texas Reading Academies. Panels (c) and (d) show event study estimates for the same outcomes, with the reference year (2024–25) indicated by a hollow marker. Texas Reading Academies were first implemented in the 2025–26 school year. Pass rate is measured as the proportion of first-attempt test takers who pass the exam (score $\geq$ 240). Average score represents the mean exam score among all first-attempt test takers, standardized to have mean zero and standard deviation of one within each year. † Certification exam data through March 2026.

Table A12: Effect of Texas Reading Academies on Certification Exam Outcomes, 2025-26 Cohort

	SOR		PPR	
	Pass Rate (1)	Score (2)	Pass Rate (3)	Score (4)
Panel A. Event Study				
2020–21 × Texas Reading Academy	0.006		−0.043	−0.069
	(0.033)		(0.028)	(0.084)
2021–22 × Texas Reading Academy	0.083**	0.256**	−0.043	−0.082
	(0.028)	(0.08)	(0.032)	(0.087)
2022–23 × Texas Reading Academy	0.057	0.145	−0.038+	−0.04
	(0.038)	(0.109)	(0.022)	(0.048)
2023–24 × Texas Reading Academy	0.042	0.084	−0.056*	−0.019
	(0.025)	(0.081)	(0.023)	(0.039)
2024–25 × Texas Reading Academy	0	0	0	0
	(.)	(.)	(.)	(.)
2025–26 × Texas Reading Academy†	−0.025+	−0.06	−0.043**	−0.117**
	(0.013)	(0.055)	(0.015)	(0.043)
Panel B. Difference-in-Differences				
Post × Texas Reading Academy	−0.052*	−0.146*	−0.007	−0.075
	(0.021)	(0.062)	(0.017)	(0.049)
Joint F-test of pre-period	<0.001	<0.001	0.359	0.713
N	61,423	40,842	97,389	97,389
Pre-period mean, non-Academy	0.856	0.051	0.854	-0.184
Pre-period mean, Academy	0.865	0.058	0.815	-0.272

Notes: Table reports estimates from event study (Panel A) and difference-in-differences (Panel B) specifications comparing certification exam outcomes at Educator Preparation Programs (EPPs) that adopted Texas Reading Academies in 2025–26 to EPPs that never adopted. Regressions are estimated on first-attempt exam records, so EPPs are weighted by their number of test takers. The outcomes are an indicator for passing on the first attempt and the standardized first-attempt score on the SOR exam (the TExES Science of Teaching Reading exam, exam 293) and the Pedagogy and Professional Responsibilities (PPR) exam. The PPR sample includes all first-attempt PPR takers at each EPP, including candidates in certification fields that do not require the SOR exam. All specifications include EPP fixed effects and year fixed effects. The reference year for the event study is 2024–25, the year immediately preceding academy implementation for the second cohort. The joint F-test reports the p-value from a test that all pre-period interaction coefficients are jointly zero. EPPs that adopted Texas Reading Academies in 2024–25 are excluded from both groups. † Certification exam data through March 1, 2026. Standard errors clustered at the Educator Preparation Program level in parentheses.

$^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Appendix B. Texas Teacher Certification Exam Descriptions

This appendix provides an overview of the Pedagogy and Professional Responsibilities (PPR) and Science of Reading (SOR) Texas teacher certification exams referenced in this study, including their domain structures and representative sample items.

Pedagogy and Professional Responsibilities EC–12 (Test 160)

The Pedagogy and Professional Responsibilities (PPR) EC–12 exam assesses candidates' knowledge of instructional design, classroom management, and professional conduct. The exam is organized into four domains:

1. **Domain I: Designing Instruction and Assessment to Promote Student Learning.** This domain addresses human developmental processes, learner diversity, instructional planning, and factors that impact student learning. Topics include developmentally appropriate practice, differentiated instruction, curriculum implementation, and critical thinking.

2. **Domain II: Creating a Positive, Productive Classroom Environment.** This domain covers establishing a safe and productive classroom climate, organizing the learning environment, and managing student behavior. Topics include physical classroom arrangement, fostering a passion for learning, formative monitoring, and cooperative group work.

3. **Domain III: Implementing Effective, Responsive Instruction and Assessment.** This domain focuses on communication strategies, active student engagement, technology integration, and monitoring student performance. Topics include questioning techniques, English-language learner supports, BYOD policies, and providing quality feedback.

4. **Domain IV: Fulfilling Professional Roles and Responsibilities.** This domain addresses family involvement, professional development, and legal and ethical requirements for Texas educators. Topics include parent partnerships, mentoring, vertical and horizontal teaming, mandatory reporting of suspected abuse, and IEP development.

Example items:

> *Which of the following activities would be the most developmentally appropriate for a first-grade teacher to include in a unit on the life cycle of plants?*
> (A) Planting a bean and observing the stages of growth
> (B) Reading nonfiction books about plants and creating posters
> (C) Visiting a science center and touring the plant exhibits
> (D) Watching a video about plants and drawing the bean life cycle

Which of the following instructional practices best ensures that a teacher structures the learning environment to address the needs of all students?
(A) Differentiation (B) Remediation (C) Enrichment (D) Modeling

Which of the following is the most appropriate first step to take if a teacher is concerned that a student is suffering from abuse?
(A) Calling the parents for a conference to discuss the suspicions
(B) Reporting the suspicions to the child protective services agency
(C) Sharing the concerns with the designated school personnel
(D) Asking the student pointed questions to ensure that all is well at home

Science of Teaching Reading (Test 293)

The Science of Teaching Reading (STR) exam assesses candidates' knowledge of evidence-based reading instruction for early childhood through grade six. The exam is organized into three domains:

1. **Domain I: Reading Pedagogy.** This domain covers foundational concepts in the science of reading, including assets-based approaches, dyslexia intervention, and reading assessment. Topics include systematic explicit instruction, formative and diagnostic assessment, and prevention of reading difficulties.

2. **Domain II: Reading Development—Foundational Skills.** This domain addresses phonological and phonemic awareness, print concepts and alphabet knowledge, phonics and word identification, syllabication and morphemic analysis, and reading fluency. Topics include phoneme-grapheme mapping, the alphabetic principle, decodable texts, inflectional endings, prosody, and automaticity.

3. **Domain III: Reading Development—Comprehension.** This domain covers vocabulary development (including Tier Two and Tier Three words), reading comprehension strategies, comprehension of literary texts, and comprehension of informational texts. Topics include background knowledge activation, text structure, metacognitive strategies, and scaffolding for English learners.

Example items:

A first-grade student has been identified as having dyslexia and has begun intervention. Which of the following approaches to instruction would be most effective to enhance the student's reading development?
(A) Allowing the student to use colored overlays on all classroom texts
(B) Using reading materials with specialized fonts designed for people with dyslexia
(C) Arranging for the student to use a working-memory training program daily
(D) Providing systematic, explicit multimodal instruction in all essential, evidence-based components of reading

Which of the following sets of words would be most appropriate to categorize as Tier Two words?
(A) fossil, air, soil (B) clock, book, floor
(C) arrange, observe, predict (D) custom, shelter, community

A third-grade English learner has grade-level decoding skills and scores around the grade-level benchmark for oral reading fluency. However, the student's text comprehension is mixed. When the student is having difficulty with a text, the teacher's best initial response should be to:
(A) Provide the student with a list of probing questions to answer after reading
(B) Talk with the student to informally assess background knowledge about the topic
(C) Advise the student to read more slowly and focus on comprehension
(D) Encourage the student to develop a written summary while reading

Score Reporting to Candidates

Candidates receive TExES score reports through their Pearson account beginning at 10:00pm Central time on the score report date, typically within 7–10 business days for selected-response exams; exams with constructed-response components take longer because trained scorers grade those responses. Figure B1 presents an example report for the English Language Arts and Reading 7–12 exam (Test 331). The report displays the candidate's total scaled score, pass/not-pass status, the scaled score range (100–300), and the minimum passing score (240). It also provides a performance breakdown by domain and competency, reporting points earned out of points possible for each, with constructed-response items reported separately. Under Texas Education Code, candidates are limited to five attempts per exam, but may appeal this with the Texas Education Agency.

Texas
Educator Certification
Examination Program

This barcode contains unique candidate information for security purposes.

Examinee Score Report

Test: 331 ENG LANGUAGE ARTS AND READING 7-12

Total Scaled Score: 262
Status: Passed
Scaled Score Range: 100-300
Passing Score: 240

Test Date: MM/DD/YYYY
FIRSTNAME M LASTNAME
123 SAMPLE STREET
CITY, TX 99999

TEA ID: 9999999

Performance by Domain and Competency	Points Possible	Points Earned
I. Reading Instruction and Assessment	23	14
001 Foundations of Reading Instruction and Assessment	8	4
002 Vocabulary Development	8	5
003 Reading Comprehension	7	5
II. Text Comprehension and Analysis	14	8
004 Reading Literary Texts	7	3
005 Reading Informational and Argumentative Texts	7	5
III. Oral and Written Communication	23	12
006 Composition	8	3
007 Inquiry and Research	8	4
008 Listening and Speaking	7	5
IV. Educating All Learners and Professional Practice	12	10
009 Differentiation Strategies in Planning and Practice	4	3
010 Culturally Responsive Practices	4	3
011 Data-Driven Practice and Formal/Informal Assessment	4	4
V. Constructed Response	8	8

Your results for the constructed-response item are reported above. For more information about the constructed-response item, see the preparation manual for this exam available on the testing program website at www.tx.nesinc.com.

See next page for additional information.

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Figure B1: Example TExES Score Report

Notes: This figure shows an example score report for the TExES English Language Arts and Reading 7–12 (Test 331) exam. The report displays the candidate's Total Scaled Score, pass/not-pass status, the scaled score range (100–300), and the minimum passing score (240), followed by a Performance by Domain and Competency breakdown reporting points earned out of points possible for each domain and competency. Source: Texas Educator Certification Examination Program (www.tx.nesinc.com).